FARM DIGITAL TWIN

Master Technical Reference

Architecture, scientific models, every formula, data pipeline, agentic AI, optimisation, calibration and build plan

A probabilistic, agent-assisted digital twin for small farm properties: soil, water, crop, solar, agrivoltaics, finance.

Pilot context: arid NW India (Jodhpur-type climate), three plots of 800 / 600 / 1300 sq ft (74.3 / 55.7 / 120.8 m²).

Companion to FarmDigitalTwin_Technical_Blueprint.zip (code-skeleton specs for Codex).

Source inspection date: 2026-11-07 · Document generated 2026-11-08

Contents

0. How to read this document 5

0.1 Evidence legend (used throughout) . . . 5

0.2 Notation . . . 5

1. Purpose, scope and success criteria 6

1.1 Pilot facts that shape the design . . . 6

1.2 Non-goals . . . 6

1.3 Success criteria (testable) . . . 6

2. System architecture 7

2.1 Layered view . . . 7

2.2 End-to-end data flow . . . 7

2.3 The daily operator-splitting step . . . 8

2.4 Coupling registry (what feeds what) . . . 8

2.5 Fidelity levels . . . 8

2.6 Reproducibility record . . . 8

3. Domain model, provenance and units 9

3.1 Entities . . . 9

3.2 Five-way provenance . . . 9

3.3 Units and boundary conversions . . . 9

3.4 Database (PostgreSQL 16 + PostGIS) . . . 10

4. Geospatial and satellite layer 10

4.1 Local CRS and area . . . 10

4.2 Terrain (Horn 1981) . . . 10

4.3 Sentinel-2 indices and the mixed-pixel policy . . . 10

5. Weather and reference evapotranspiration 11

5.1 Weather sources and blending . . . 11

5.2 FAO-56 Penman–Monteith (daily) . . . 11

6. Soil hydraulics and the layered water balance 12

6.1 Pedotransfer functions (when θwp,fc,s and Ksat are not measured) . . . 12

6.2 Daily water balance — process order . . . 12

7. Crop engine (Level 1, water-driven) 14

7.1 Daily order of operations . . . 14

7.2 Phenology and roots . . . 14

7.3 Canopy cover . . . 14

7.4 Water stress coefficients . . . 15

7.5 Transpiration and biomass . . . 15
7.6 Harvest index and yield . . . 15
7.7 Crop library and placeholder policy . . . 15
7.8 Detailed-model adapter (Level 3) . . . 16
7.9 Golden regression values (generated by running the reference model) . . . 16

8. Salinity and sodicity (original module) 17

8.1 Units and initial condition . . . 17
8.2 Daily salt step (after the water balance) . . . 17
8.3 Crop response and sodicity . . . 17
8.4 Leaching requirement and reclamation . . . 17

9. Soil fertility: nitrogen, phosphorus, potassium 18

9.1 Nitrogen pools and fluxes (per layer, daily) . . . 18
9.2 Crop demand, uptake and stress . . . 18
9.3 Phosphorus, potassium, micronutrients, recommendations . . . 18
9.4 Amendments and organic inputs . . . 18

10. Soil organic carbon (RothC-type, monthly) 19

10.1 Inputs, initialisation, outputs . . . 19

11. Solar PV energy model (hourly) 20

11.1 Solar position and extraterrestrial radiation . . . 20
11.2 Irradiance decomposition (Erbs) . . . 20
11.3 Transposition to the plane of array . . . 20
11.4 Soiling, cell temperature, DC and AC power . . . 20
11.5 Layout and sizing for tiny plots . . . 21
11.6 Golden clear-sky values (Jodhpur, 1 kWp) . . . 21

12. Agrivoltaics: ground radiation and crop coupling 22

12.1 Geometry (infinite rows, cross-section plane) . . . 22
12.2 Diffuse sky and ground irradiance . . . 22
12.3 Crop coupling mechanisms (each switchable) . . . 22

13. Finance and economics 23

13.1 Core metrics . . . 23
13.2 Crop budget, PV project, agrivoltaic combination . . . 23
13.3 Water and energy costs . . . 24

14. Uncertainty quantification 24

14.1 Weather scenario generation . . . 24
14.2 Propagation and variance attribution . . . 24
14.3 Global sensitivity analysis . . . 24

15. Calibration, validation and safe learning **25**

15.1 Four-level hierarchy with partial pooling . . . 25
15.2 Inference methods . . . 25
15.3 Scoring . . . 25
15.4 The promotion gate (no candidate is used unless it proves itself) . . . 25
15.5 Residual machine learning (only after enough data) . . . 26
15.6 Closing the loop: error attribution . . . 26

16. Optimisation and value of information **27**

16.1 Decision structure . . . 27
16.2 Risk-aware MILP (Rockafellar–Uryasev CVaR) . . . 27
16.3 Value of information . . . 27

17. Data ingestion and quality **28**

17.1 Soil-report ingestion pipeline . . . 28
17.2 Data-quality engine . . . 28

18. Agentic AI layer **29**

18.1 Shared state and evidence ledger . . . 29
18.2 Typed tool layer . . . 29
18.3 Graph behaviour . . . 30
18.4 Governance, permissions and prohibitions . . . 30
18.5 Agent evaluation suite (CI) . . . 30

19. API, UI and security **31**

19.1 API (FastAPI, /api/v1) . . . 31
19.2 UI (Streamlit first; mobile-first, English + Hindi) . . . 31
19.3 Security and privacy . . . 31

20. Verified vs. unverified: the honesty ledger **32**

20.1 What was actually inspected . . . 32
20.2 Findings that changed the design . . . 32
20.3 What Codex must verify before relying on it . . . 32
20.4 Risk register . . . 32

21. Build plan, testing and acceptance **33**

21.1 Test families . . . 33
21.2 Field-validation plan . . . 33
21.3 Data to collect first (typical value order) . . . 33

22. Appendix A — Glossary **34**

23. Appendix B — Code-skeleton file index **34**

0. How to read this document

This PDF is the human-readable companion to the code-skeleton ZIP. The ZIP tells Codex *what to code* file by file; this document explains *why* the system is shaped the way it is and states **every formula** the models use, in typeset form, with units, worked numbers and the evidence status of each one.

0.1 Evidence legend (used throughout)

VERIFIED IN SOURCE. The equation or constant was read from real source code during blueprint creation (AquaCrop-OSPy, pyRothC, pvlib, DSSAT tree, farmOS docs, LangGraph) or reproduced by running software.

LITERATURE / UNVERIFIED HERE. Written from domain knowledge of the published method. Not re-checked against the paper in this sandbox. Codex must verify before use and log the check.

DESIGN DECISION. A design choice of this project (not a fact about nature). Alternatives are listed where relevant.

WARNING. A known limitation that must be shown to users with the result.

0.2 Notation

Symbol	Meaning	Unit
θ, θ_{wp}, θ_{fc}, θ_{s}	volumetric water content; wilting point; field capacity; saturation	m^3/m^3
Dr, TAW	root-zone depletion; total available water	mm
ET_0, Tr, Es	reference evapotranspiration; crop transpiration; soil evaporation	mm/day
CC	canopy cover (fraction of ground covered)	0–1
K_s	stress coefficient (1 = no stress)	0–1
B, HI, Y	biomass; harvest index; yield	t/ha, –, t/ha
EC_e, EC_{iw}	saturated-paste EC; irrigation-water EC	dS/m
SAR, ESP	sodium adsorption ratio; exchangeable sodium percentage	$(mmol/L)^{½}$, %
TOC, C	total organic carbon; carbon pool	t C/ha
G, POA, GHI, DNI, DHI	irradiance (plane-of-array, global horizontal, direct normal, diffuse horizontal)	W/m^2
NPV, $CVaR_{\alpha}$	net present value; conditional value-at-risk at level α	INR

1. Purpose, scope and success criteria

The system answers one family of questions for a specific property: *given my soil, water, climate, capital and goals, what should each plot be used for, which crops, how should it be irrigated and amended, should I put solar or agrivoltaics on any of it, and how risky is each choice?* It does this by keeping an honest, versioned model of the property (the **FarmState**), simulating futures under uncertainty, optimising decisions under risk, and learning from outcomes — but only when learning provably helps.

1.1 Pilot facts that shape the design

Fact	Consequence for design
Plots are 55–120 m² (smaller than the area of 1.2 Sentinel-2 pixels)	Mixed-pixel policy for satellite data; never present a blended pixel as plot-pure; wide uncertainty on satellite-derived cover.
Yield per plot is kilograms, PV is single-digit kWp	Economics are dominated by water, labour and capex — not field-scale yield; unit tests use small numbers.
Arid climate, likely saline/brackish groundwater	Salinity/sodicity and leaching are first-class models (the reference crop engine has none).
Indian arid crops (pearl millet, cluster bean, mustard, chickpea…) are absent from the reference crop library	Placeholder parameters with visible flags; local calibration; optional detailed-model cross-check.
Plot C may host PV / agrivoltaics	Original ground-radiation geometry and an experimental shade-response model, clearly labelled unvalidated.

1.2 Non-goals

- No electrical or structural design sign-off (licensed professionals required).
- No carbon-credit verification (carbon is reported as tC and tCO_2e for decision support only).
- No automatic purchases or payments; no pest/disease prediction at Level 1; no perennial/tree crops at Level 1 (raises NotImplementedError).

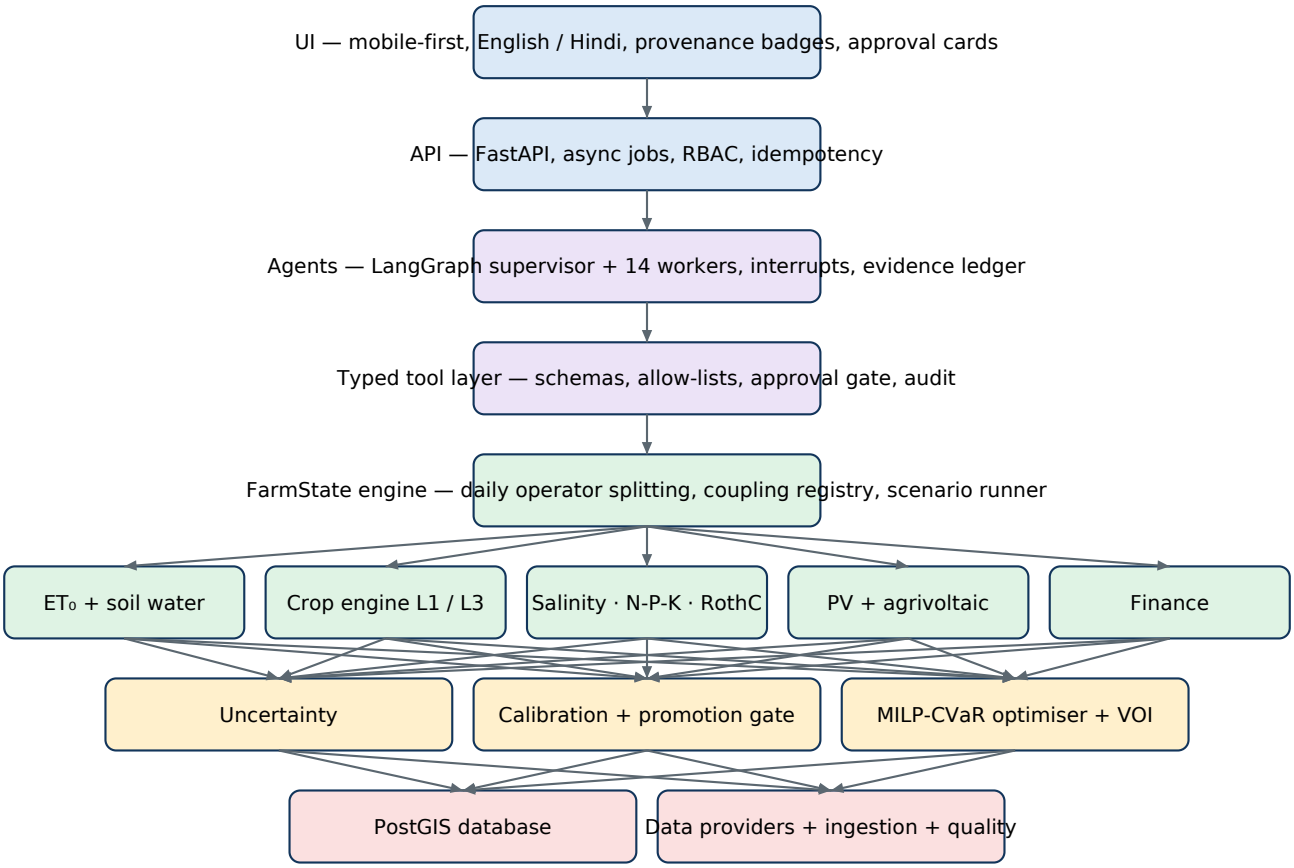
1.3 Success criteria (testable)

Criterion	Target
Reference crop regression (Tunis wheat)	yield ±3 %, seasonal irrigation ±5 mm, daily canopy cover ±0.01
Mass balances (water, salt, N, C)	water 0.5 mm/season; salt 1 kg/ha; N 0.5 kg/ha; C 1e-9 relative
PV golden (clear-sky Jodhpur)	GHI/POA ±2 %, DC/AC ±3 %, solar zenith ±0.1°
Calibration safety	candidate must beat parent on blocked held-out data; automatic rollback on drift
Agent grounding	≥ 98 % of stated numbers trace to tool evidence; 100 % refusal on forbidden actions
Performance	1 plot × 1 season × 100 scenarios < 1 s; 10 yr × 50 scen × 5 plots < 5 min (CPU only)

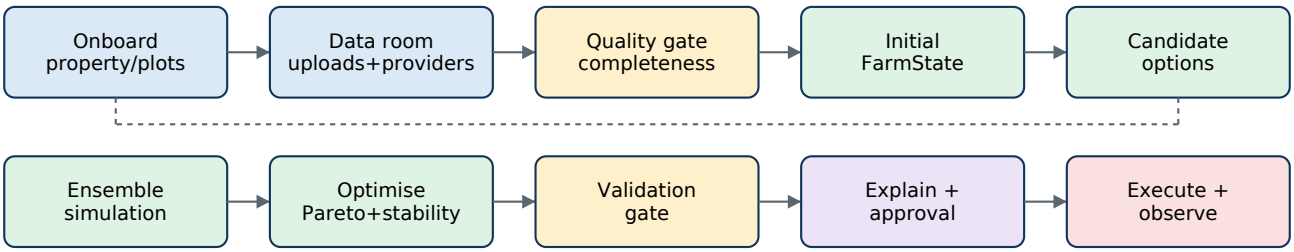
2. System architecture

The architecture rests on six principles. (1) **Agents orchestrate; kernels compute** — language models never produce numbers; deterministic tools do. (2) **One state, many models** — every model reads and writes the same FarmState through Protocol interfaces and can be swapped. (3) **Fidelity levels** — a simple native model is always available (L1), richer variants need more data (L2), and an external detailed process model is used only for cross-checks (L3). (4) **Explicit coupling** with a one-day lag to break feedback cycles. (5) **Uncertainty is first-class** and decomposed into weather, parameter and economic parts. (6) **Learning is gated** and reversible.

2.1 Layered view



2.2 End-to-end data flow



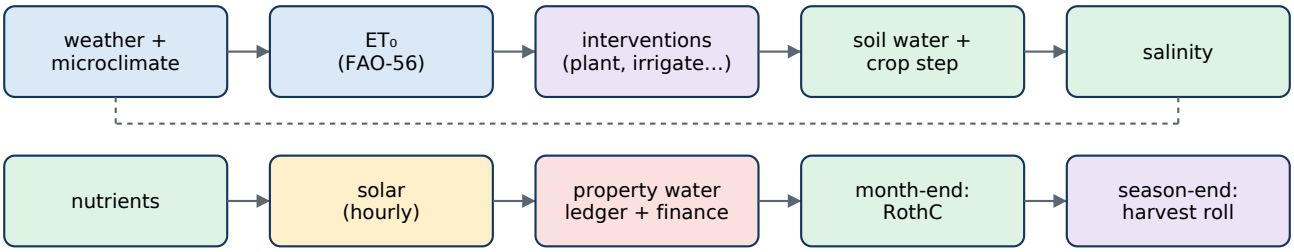
- **Onboarding**: user → Intake agent → Property/Plot (area computed in a local metric CRS, never in degrees).
- **Data room**: uploads and providers → parsing → validation → `observation` (measured/imported) and `estimate` (pedotransfer, priors) tables; quality report and per-model completeness.
- **Quality gate**: if completeness is too low, the system asks for the VOI-ranked most useful measurement, or the user may explicitly “proceed with assumptions” and every output then carries

an assumptions banner.

- **Simulation**: weather × parameter draws × price paths run through the engine to fill a payoff table (NPV, water, carbon, EC_e, capex per plot-option-scenario).
- **Optimisation**: linear risk-aware model over the payoff table; Pareto frontier; decision stability by bootstrap.
- **Close the loop**: outcome → error attribution (weather vs management vs model) → calibration candidate → promotion gate → active parameters.

2.3 The daily operator-splitting step

For each day and plot the engine applies the same ordered sequence. The order matters for mass conservation. Water–salinity–nutrient–crop feedbacks are cut by a **one-day lag**: stress factors computed today act tomorrow.



2.4 Coupling registry (what feeds what)

Source	Output	Target	Lag
weather	Tmax, Tmin, P, Rs, RH, u_2	ET_0 module	0
ET_0	daily mm	crop engine, evaporation	0
microclimate	frad, u_2 scale, ΔT	ET_0, crop, soil temperature	0
soil water	drainage / evaporation fluxes	salinity, nitrogen	0
salinity	root-zone EC_e	crop $K_{s,sal}$	**1 day**
nitrogen	$K_{s,N}$, $K_{s,PK}$	crop biomass increment	**1 day**
crop	residue carbon	RothC inputs	monthly
RothC	SOC → organic matter %	pedotransfer (annual refresh)	annual
crop	irrigation applied, yield	water ledger, finance	0
solar	kWh	finance	0
ground radiation	frad, PAR	crop coupling, ET_0, RothC	0

Capability negotiation: before a run, the engine asks each model what processes it supports. If a scenario needs salinity but the chosen crop model lacks it, a separate salinity model is composed in and the composition is recorded in the run record. A missing process is *never* silently ignored.

2.5 Fidelity levels

Level	Meaning	When used
L1	native, simple, always available	default for every process
L2	enhanced; needs more data (measured retention curves, anisotropic sky, finite rows...)	when data exist
L3	external detailed process model (DSSAT) run as a subprocess	cross-validation only; disagreement > 25 % yield raises a flag, never an automatic engine switch

2.6 Reproducibility record

Every simulation writes one `sim_run` row: configuration JSON and its SHA-256 hash, input hash, RNG seed, model versions, parameter-set ids, code git SHA. Given (input hash, seed) the engine is idempotent. Stochastic functions take an explicit `numpy.random.Generator`; there are no global seeds.

3. Domain model, provenance and units

3.1 Entities

Entity	Holds
Property	reference point, local UTM EPSG, currency, per-property regional land-unit factors (bigha etc. vary by region and are never hard-coded)
Plot	polygon (WGS-84), area in m² (from local CRS), eligible land uses, slope/aspect/elevation (each with provenance)
SoilState / SoilLayer	per-layer texture, density, OM/OC, $\theta_{wp,fc,s}$, K_{sat}, τ, pH, EC_e, SAR/ESP, CEC, N, P, K, micronutrients, salt mass, RothC pools; evaporation state
CropState	dap, GDD, CC, root depth, biomass, HI factors, stress counters, N uptake, status
WaterState	sources (well, tank, canal, tanker), storages, daily ledger
SolarState / PVConfig	kWp, tilt, azimuth, pitch, clearance, losses, degradation, soiling state
FinancialState	budget, price series, cost catalogue, cashflow ledger (ACTUAL vs SIMULATED never mixed), tariff schedule
FarmState	date + property state + plot states + weather today + metadata; snapshot / diff / hash / validate
Intervention	Plant, Harvest, Irrigate, IrrigationPolicy, ApplyFertiliser, ApplyOrganic, ApplyAmendment, ChangeLandUse, InstallPV, Mulch…
Scenario	base state + plan + weather/price spec + parameter sets + horizon + seed

3.2 Five-way provenance

Every value inside the FarmState is a `Quantified` object: *(value, unit, provenance)*, with provenance source in {MEASURED, IMPORTED, ESTIMATED, MODEL_PREDICTED, MISSING}. Rules that code must enforce:

- ESTIMATED requires a method, parent ids and an uncertainty; MODEL_PREDICTED requires model version and parameter-set id.
- MEASURED can never be overwritten by a non-measured value; corrections create a new row that supersedes the old one.
- MISSING renders as a gap in the UI — never as zero — and cannot be silently filled; it may only be replaced by an ESTIMATED value that carries its method and uncertainty.
- Database: `observation` (MEASURED/IMPORTED only), `estimate`, and `prediction` are physically separate tables; triggers forbid MODEL_PREDICTED in observation.

3.3 Units and boundary conversions

Internally: mm, mm/day, m, m³/m³, °C, MJ/m²/day (daily) or W/m² (hourly), m², t/ha, kg/ha, INR, kWh, dS/m. Conversions happen only at I/O boundaries.

$$1\ \text{sq ft} = 0.09290304\ \text{m}^2 \quad \Rightarrow \quad 800,\ 600,\ 1300\ \text{sq ft} = 74.32,\ 55.74,\ 120.77\ \text{m}^2 \tag{E3.1}$$

$$m_{\text{kg/ha}} = c_{\text{mg/kg}} \cdot \rho_b\,[\text{g/cm}^3] \cdot d\,[\text{m}] \cdot 10 \qquad (\text{e.g. } 10 \cdot 1.4 \cdot 0.15 \cdot 10 = 21\ \text{kg/ha}) \tag{E3.2}$$

$$V_{\text{m}^3} = \frac{h_{\text{mm}} \cdot A_{\text{m}^2}}{1000} \tag{E3.3}$$

$$\text{OM}\,[\%] = 1.724 \cdot \text{OC}\,[\%] \quad (\text{Van Bemmelen, configurable}) \tag{E3.4}$$

$$\text{meq/L} = \frac{\text{mg/L}}{\text{equivalent weight}}; \quad \text{Ca } 20.04,\ \text{Mg } 12.15,\ \text{Na } 22.99,\ \text{K } 39.10 \tag{E3.5}$$

Range validation (RangeError on violation): pH 3.5–10.5; EC 0–60 dS/m; OC 0–12 %; sand+silt+clay = 100 ± 2; bulk density 0.8–1.9 g/cm³; θ 0–0.7; air temperature −40…60 °C; rainfall 0–1000 mm/day; ET_0 0–15 mm/day; R_s 0–40 MJ/m²/day.

3.4 Database (PostgreSQL 16 + PostGIS)

Table	Purpose
property, plot	spatial skeleton (geometry + area CHECK > 0, GiST index)
variable_def	registry of variables, canonical units, valid ranges
dataset	every upload/provider pull with SHA-256, licence, extent
observation / estimate / prediction	MEASURED-IMPORTED / ESTIMATED / MODEL_PREDICTED, separately
parameter_set	versioned parameters at GENERIC/REGIONAL/FARM/PLOT with status CANDIDATE → VALIDATED → ACTIVE → RETIRED / REJECTED
sim_run, state_delta	reproducibility record; state stored as deltas + monthly snapshots
recommendation, approval, intervention, outcome	the decision-to-outcome chain
calibration_run	train/holdout indices, metrics before/after, decision, reason
agent_run, tool_call, audit_log	full agent trace; audit is append-only (trigger forbids UPDATE/DELETE)
weather_daily, price_observation, sat_scene	cached provider data with provenance

4. Geospatial and satellite layer

4.1 Local CRS and area

Plot area is computed after transforming to a local UTM zone (pyproj/shapely) — never from degrees. The UTM zone and EPSG code follow:

$$\text{zone} = \left\lfloor \frac{\lambda + 180}{6} \right\rfloor + 1, \qquad \text{EPSG} = 32600 + \text{zone (N)} \;\; \text{or} \;\; 32700 + \text{zone (S)} \tag{E4.1}$$

Jodhpur (73.02°E, 26.24°N): zone = ⌊(73.02+180)/6⌋+1 = 43 → EPSG 32643. Plot entry modes: map drawing, GeoJSON/KML/Shapefile/GeoPackage import, or dimensions (rectangle L×W with bearing, or vertex list in metres). Validation: polygon valid and simple, area within ±2 % of user-entered dimensions, overlap with siblings < 0.5 %.

4.2 Terrain (Horn 1981)

For a 3×3 elevation window with cells a b c / d e f / g h i (row-major, a = top-left), spacing Δx, Δy:

$$\frac{\partial z}{\partial x} = \frac{(c + 2f + i) - (a + 2d + g)}{8\,\Delta x}, \qquad \frac{\partial z}{\partial y} = \frac{(g + 2h + i) - (a + 2b + c)}{8\,\Delta y} \tag{E4.2}$$

$$\text{slope} = \arctan\sqrt{(\partial z/\partial x)^2 + (\partial z/\partial y)^2}, \qquad \text{aspect} = \text{atan2}(\partial z/\partial y,\; -\partial z/\partial x) \;\rightarrow\; \text{compass} \tag{E4.3}$$

WARNING. A 30 m DEM is sub-pixel for a 100 m² plot. DEM-derived slope/aspect/elevation are ESTIMATED with DEM vertical error as uncertainty; a phone-clinometer or survey value overrides as MEASURED.

4.3 Sentinel-2 indices and the mixed-pixel policy

$$\text{NDVI} = \frac{B8 - B4}{B8 + B4}, \quad \text{EVI} = 2.5\,\frac{B8 - B4}{B8 + 6B4 - 7.5B2 + 1}, \quad \text{NDWI} = \frac{B3 - B8}{B3 + B8}, \quad \text{NDMI} = \frac{B8 - B11}{B8 + B11} \tag{E4.4}$$

$$\text{SAVI} = 1.5\,\frac{B8 - B4}{B8 + B4 + 0.5}, \qquad f_{\text{cover}} = \left(\frac{\text{NDVI} - \text{NDVI}_{soil}}{\text{NDVI}_{veg} - \text{NDVI}_{soil}}\right)^2 \;\; (0.15,\; 0.85) \tag{E4.5}$$

L2A digital numbers are divided by 10 000, after applying the BOA additive offset *read from product metadata* (processing baseline ≥ 04.00 uses −1000) — never hard-coded. Cloud mask keeps SCL classes {4 vegetation, 5 bare soil, 6 water} and drops {0, 1, 3, 8, 9, 10, 11}.

Mixed-pixel policy. A 74 m² plot covers less than one 100 m² pixel. The plot statistic is an area-weighted mean of intersecting pixels, with weight = intersection area ÷ pixel area. The system reports n_{pure} (pixels ≥ 95 % inside the plot); if n_{pure} = 0 the quality flag `sub_pixel_plot` is set and uncertainty is inflated. Time series are regularised to a 5-day grid and smoothed (Savitzky–Golay after linear interpolation, or double-logistic fit), with raw and smoothed both kept. Anomaly rule: z-score against the plot's own multi-year phenology envelope; flag NDVI drops > 0.15 within 10 days.

5. Weather and reference evapotranspiration

5.1 Weather sources and blending

Source	Role	Trust / provenance
Own station / CSV upload	ground truth	MEASURED
IMD gridded rain/temperature (file-based loader)	best India rainfall for calibration	IMPORTED
NASA POWER	long climatology 1981–present; weather-generator fitting	IMPORTED, gridded
Open-Meteo archive/forecast	forecast + reanalysis	IMPORTED, coarse (~10–25 km) — flag gridded_reanalysis

Priority per variable: MEASURED station > IMD (rain) > NASA POWER / Open-Meteo. Where station and gridded data overlap, a bias correction (monthly multiplicative for rain, additive for temperature) is computed and stored as ESTIMATED with uncertainty. Provider ET_0 is only a cross-check; the system computes its own.

LITERATURE / UNVERIFIED HERE. All endpoint URLs, parameter names and units for providers were written from memory; the authoring sandbox had restricted internet access. Codex must verify each one live (NASA POWER reports missing values as −999 → convert to MISSING; read radiation units from the response metadata).

5.2 FAO-56 Penman–Monteith (daily)

$$ET_0 = \frac{0.408\,\Delta\,(R_n - G) + \gamma\,\frac{900}{T+273}\,u_2\,(e_s - e_a)}{\Delta + \gamma\,(1 + 0.34\,u_2)} \qquad \text{E5.1}$$

with G = 0 for a daily step and T = (T_{max}+T_{min})/2. Supporting equations:

$$e^\circ(T) = 0.6108\exp\!\left(\frac{17.27\,T}{T+237.3}\right), \qquad e_s = \frac{e^\circ(T_{max}) + e^\circ(T_{min})}{2} \qquad \text{E5.2}$$

$$\Delta = \frac{4098\,e^\circ(T)}{(T+237.3)^2}, \qquad P = 101.3\left(\frac{293 - 0.0065\,z}{293}\right)^{5.26}, \qquad \gamma = 0.000665\,P \qquad \text{E5.3}$$

$$e_a = e^\circ(T_{dew}) \;\text{ or }\; e_a = \frac{RH_{mean}}{100}\,e_s \;\text{ or }\; e^\circ(T_{min} - K_0) \quad (K_0 = 2\text{ C};\ 4\text{ C in arid zones}) \qquad \text{E5.4}$$

$$d_r = 1 + 0.033\cos\!\left(\frac{2\pi J}{365}\right), \quad \delta = 0.409\sin\!\left(\frac{2\pi J}{365} - 1.39\right), \quad \omega_s = \arccos(-\tan\varphi\tan\delta) \qquad \text{E5.5}$$

$$R_a = \frac{24\cdot 60}{\pi}\,G_{sc}\,d_r\,[\omega_s\sin\varphi\sin\delta + \cos\varphi\cos\delta\sin\omega_s], \quad G_{sc} = 0.0820\ \text{MJ m}^{-2}\,\text{min}^{-1} \qquad \text{E5.6}$$

$$R_{so} = (0.75 + 2\times10^{-5}z)\,R_a, \quad R_{ns} = 0.77\,R_s, \quad R_{nl} = \sigma\,\frac{T_{max,K}^4 + T_{min,K}^4}{2}\,(0.34 - 0.14\sqrt{e_a})\,(1.35\,\frac{R_s}{R_{so}} - 0.35) \qquad \text{E5.7}$$

$$R_n = R_{ns} - R_{nl}, \qquad \sigma = 4.903\times10^{-9}\ \text{MJ K}^{-4}\,\text{m}^{-2}\,\text{day}^{-1} \quad (R_s/R_{so}\ \text{clipped to}\ [0.3, 1]) \qquad \text{E5.8}$$

$$u_2 = u_z\,\frac{4.87}{\ln(67.8\,z_m - 5.42)} \qquad (u_{10} = 3.0 \Rightarrow u_2 = 2.24\ \text{m/s}) \qquad \text{E5.9}$$

Fallbacks (each flagged ESTIMATED with wider uncertainty)

$$R_s = k_{Rs}\sqrt{T_{max} - T_{min}}\;R_a \quad (k_{Rs} = 0.16\ \text{interior},\ 0.19\ \text{coastal}) \qquad \text{E5.10}$$

$$ET_0^{\mathrm{Har}} = 0.0023 \cdot 0.408\, R_a\, (T_{mean} + 17.8)\sqrt{T_{max} - T_{min}} \qquad (\pm 15\%\ \text{uncertainty}) \qquad \text{E5.11}$$

Microclimate variant (used by agrivoltaics): the same equation is re-run with R_s multiplied by the ground-radiation fraction f_{rad}, u_2 multiplied by a wind factor, and T_{max}, T_{min} shifted by ΔT. The radiation, wind and temperature effects are independently switchable for sensitivity analysis. Test: FAO-56 worked daily example (Uccle) must reproduce $ET_0 \approx 3.9$ mm/day within 0.05.

6. Soil hydraulics and the layered water balance

6.1 Pedotransfer functions (when $\theta_{wp,fc,s}$ and K_{sat} are not measured)

Regressions of Saxton–Rawls type, with S = sand fraction (0–1), C = clay fraction (0–1), OM = organic matter (%), DF = density factor (0.9–1.3, default 1.0). These are the exact forms used by the reference crop model's soil generator.

VERIFIED IN SOURCE. Coefficients below were read from the reference implementation (AquaCrop-OSPy soil entity). They are empirical: typical θ error 0.02–0.04 m³/m³, and K_{sat} uncertain by a factor of several — so outputs always carry an uncertainty (θ: sd 0.03; ln K_{sat}: sd 1.0).

$$\theta_{1500}^{t} = -0.024S + 0.487C + 0.006\,OM + 0.005\,S{\cdot}OM - 0.013\,C{\cdot}OM + 0.068\,S{\cdot}C + 0.031, \qquad \theta_{wp} = \theta_{1500}^{t} + 0.14\,\theta_{1500}^{t} - 0.02 \qquad \text{E6.1}$$

$$\theta_{33}^{t} = -0.251S + 0.195C + 0.011\,OM + 0.006\,S{\cdot}OM - 0.027\,C{\cdot}OM + 0.452\,S{\cdot}C + 0.299, \qquad \theta_{33} = \theta_{33}^{t} + (1.283\,\theta_{33}^{t2} - 0.374\,\theta_{33}^{t} - 0.015) \qquad \text{E6.2}$$

$$\theta_{S-33}^{t} = 0.278S + 0.034C + 0.022\,OM - 0.018\,S{\cdot}OM - 0.027\,C{\cdot}OM - 0.584\,S{\cdot}C + 0.078, \qquad \theta_{S-33} = \theta_{S-33}^{t} + (0.636\,\theta_{S-33}^{t} - 0.107) \qquad \text{E6.3}$$

$$\theta_s^{N} = \theta_{33} + \theta_{S-33} - 0.097S + 0.043, \qquad \rho_N = (1 - \theta_s^{N}) \cdot 2.65, \qquad \rho_{DF} = \rho_N \cdot DF \qquad \text{E6.4}$$

$$\theta_s = 1 - \frac{\rho_{DF}}{2.65}, \qquad \theta_{fc} = \theta_{33} + 0.2\left[\left(1 - \frac{\rho_{DF}}{2.65}\right) - \left(1 - \frac{\rho_N}{2.65}\right)\right] \qquad \text{E6.5}$$

$$\lambda = \frac{\ln\theta_{fc} - \ln\theta_{wp}}{\ln 1500 - \ln 33}, \qquad K_{sat} = 1930\,(\theta_s - \theta_{fc})^{\,3-\lambda} \cdot 24 \ \ [\mathrm{mm/day}], \qquad \theta_{dry} = \theta_{wp}/2 \qquad \text{E6.6}$$

$$\tau = \mathrm{round}\left(0.0866\,K_{sat}^{0.35},\ 2\right) \ \ \text{clamped to } [0,1] \qquad (\text{drainage characteristic}) \qquad \text{E6.7}$$

Worked values computed from equations E6.1–E6.7 while building this document (use them as hand-check golden values in tests):

Soil (S, C, OM)	θ_{wp}	θ_{fc}	θ_s	K_{sat} mm/day	τ
Sand-ish (S .85, C .05, OM 1)	0.034	0.087	0.426	2351	1.00
Loam (S .40, C .20, OM 2)	0.134	0.274	0.446	329	0.66
Clay loam (S .30, C .35, OM 2)	0.216	0.355	0.469	90	0.42

Capillary-rise parameters (a_{CR}, b_{CR}), readily evaporable water (REW) and curve numbers are shipped as **editable CSV tables with provenance** (not hard-coded). Level-3 option: van Genuchten θ(h) and Mualem K(θ) for a Richards-equation solver, used only where measured retention curves and sensors exist.

6.2 Daily water balance — process order

The balance runs in this fixed order each day; reordering breaks mass conservation.

#	Process	#	Process
1	water-table adjustment of θ_{fc} (if a table is present)	6	infiltration into layers, capped by K_{sat}; ponding / surface storage
2	root development (Z_r)	7	capillary rise from a water table (default OFF)
3	drainage / redistribution	8	soil evaporation (two-stage)
4	curve-number runoff and infiltration partition	9	crop transpiration extraction (root sink, stress-limited)
5	irrigation (efficiency, wetted fraction)	10	groundwater inflow → stress for next day → ledger → assert closure

Drainage (per layer)

$$d\theta = 0 \qquad (\theta \le \theta_{fc}) \tag{E6.8a}$$

$$d\theta = \tau\,(\theta_s - \theta_{fc}) \qquad (\theta \ge \theta_s) \tag{E6.8b}$$

$$d\theta = \tau\,(\theta_s - \theta_{fc})\,\frac{e^{\theta - \theta_{fc}} - 1}{e^{\theta_s - \theta_{fc}} - 1} \qquad (\theta_{fc} < \theta < \theta_s) \tag{E6.8c}$$

Flux out = dθ·Δz·1000 mm, limited so θ − dθ ≥ θ_{fc}. Layers are processed top to bottom; each layer can accept water only up to θ_s and no faster than its K_{sat}; the part it cannot accept is held back in the layer above (reduced upstream drainage). Flux out of the bottom layer is deep percolation (a loss in the ledger).

VERIFIED IN SOURCE. The drainage law and tau relation were read from the reference implementation; the "hold-back" logic must be tested on a 3-layer column.

Curve-number runoff

$$CN_{dry} = 1.4\times10^{-14} + 0.507\,CN - 0.00374\,CN^2 + 0.0000867\,CN^3, \qquad CN_{wet} = 5.6\times10^{-14} + 2.33\,CN - 0.0209\,CN^2 + 0.000076\,CN^3 \tag{E6.9}$$

$$w_x(z) = 1.016\left(1 - e^{-4.16\,z/z_{CN}}\right), \qquad w_{rel,\,i} = w_x(z_i) - w_x(z_{i-1}), \qquad W_{top} = \sum_i w_{rel,\,i}\,\frac{\theta_i - \theta_{wp,\,i}}{\theta_{fc,\,i} - \theta_{wp,\,i}} \quad (0\text{–}1) \tag{E6.10}$$

$$CN = \mathrm{round}\left(CN_{dry} + (CN_{wet} - CN_{dry})\,W_{top}\right), \quad S = \frac{25400}{CN} - 254, \quad I_a = 0.05\,S, \quad Q = \frac{(P - I_a)^2}{P + 0.95\,S} \quad (P > I_a) \tag{E6.11}$$

Worked check: P = 50 mm, CN = 75 → S = 84.67 mm, I_a = 4.23 mm, Q = 45.77²/130.43 = **16.06 mm**; infiltration = 33.94 mm. With bunds higher than 1 mm, Q = 0 and water ponds as surface storage.

Soil evaporation (two-stage, readily-evaporable-water)

$$Es_{pot} = K_{ex}\,(1 - CC_{adj})\,ET_0, \qquad Es_{pot,\,max} = K_{ex}\,ET_0\,(1 - CC_{x,\,W}\,f_{wcc}/100) \tag{E6.12}$$

$$CC_{x,\,Act,\,adj} = 1.72\,CC_{x,\,Act} - CC^2_{x,\,Act} + 0.3\,CC^3_{x,\,Act} \qquad (\text{lower bound } Es_{pot,\,min} = K_{ex}(1 - CC_{x,\,Act,\,adj})ET_0) \tag{E6.13}$$

$$\text{Stage 1:}\quad Es = Es_{pot}\ \ (\le REW); \qquad \text{Stage 2:}\quad K_r = \frac{e^{f_{evap}W_{rel}} - 1}{e^{f_{evap}} - 1}, \quad W_{rel} = \frac{W_{act} - W_{dry}}{W_{upper} - W_{dry}}, \quad Es = K_r\,Es_{pot} \tag{E6.14}$$

$$W_{upper} = W_{stage2}\,(W_{sat} - (W_{fc} - REW)) + (W_{fc} - REW) \tag{E6.15}$$

Mulch: Es_{pot} × (1 − f_{mulch}·cover/100); partial wetting after irrigation (no rain, no ponding): Es_{pot} × wetted-surface %/100, taking the smaller of the mulch- and wetting-adjusted values. The evaporating layer deepens from Z_{min} to Z_{max} in 1 mm steps when W_{rel} < $f_{WrelExp}$(Z_{max}−Z)/(Z_{max}−Z_{min}); the day is sub-stepped (default 20).

VERIFIED IN SOURCE. Defaults read from source: K_{ex} 1.1, f_{evap} 4, $f_{WrelExp}$ 0.4, f_{wcc} 50 %, REW 9.0 mm (when not texture-adjusted), Z_{min} 0.15 m, Z_{max} 0.30 m, z_{CN} 0.3 m, z_{germ} 0.3 m. Texture-based REW ranges (sand 2–7 … clay 8–12 mm) are literature (FAO-56 Table 19) and shipped as an editable CSV.

Root-zone water, depletion and irrigation

$$TAW = 1000\sum_i(\theta_{fc,\,i} - \theta_{wp,\,i})\,\Delta z_i\,\phi_i, \qquad Dr = 1000\sum_i(\theta_{fc,\,i} - \theta_i)\,\Delta z_i\,\phi_i \qquad (\phi_i = \text{root fraction in layer}) \tag{E6.16}$$

$$\text{irrigation trigger:}\quad \frac{Dr}{TAW} > 1 - \frac{SMT}{100}; \qquad \text{gross} = \frac{\text{net}}{\eta_{app}}, \quad \text{net} = Dr\ (\text{refill to FC}) \tag{E6.17}$$

Irrigation policies: fixed schedule, soil-moisture threshold per growth stage (SMT), ET-replacement, or rainfed; each with maximum depth, interval and season cap. Optional skip rule: if forecast rain over the next 48 h ≥ 0.5·Dr with probability ≥ 0.7, skip irrigation (flagged extension).

Mass-balance assertion (hard test)

$$\sum(P + Irr + CR + GW_{in}) - \sum(Q + Es + Tr + D_{deep} + D_{tile}) = \Delta S_{soil} + \Delta S_{surface} \quad (\pm 0.5\ \text{mm/season}) \tag{E6.18}$$

7. Crop engine (Level 1, water-driven)

An original implementation of published AquaCrop-style equations, validated against the reference run (Section 7.9). It is vectorised over a scenario axis. Salinity, nitrogen and light-limitation hooks are added because the reference engine lacks them.

7.1 Daily order of operations

VERIFIED IN SOURCE. Order taken from the reference daily loop; steps 9 and 13 gain extra stress terms in this project.

#	Step	#	Step
1	if outside planting–maturity: soil water only (fallow evaporation)	9	stress coefficients (water, aeration, temperature, **salinity, nitrogen**)
2	growing degree days; advance crop calendar	10	canopy development using $K_{s,exp}$ and $K_{s,sen}$
3	growth stage; crop_alive flag	11	soil evaporation with canopy shading
4	start-of-day TAW and Dr	12	transpiration demand and actual with root extraction
5	root development	13	biomass increment with all stress multipliers
6	irrigation decision → gross depth	14	harvest index / yield formation
7	drainage, runoff, infiltration, irrigation	15	state update, ledger, mass-balance check, output row
8	capillary rise / groundwater (optional)	16	maturity or death → season summary

7.2 Phenology and roots

$$GDD = \max(0,\ \overline{T} - T_{base}), \quad \overline{T} = \frac{\min(T_{max}, T_{upp}) + \max(T_{min}, T_{base})}{2} \quad \text{(method 2/3 variants selectable)} \qquad \text{E7.1}$$

$$Z_r(t) = Z_{max}\ (t \geq t_{max}), \qquad Z_r(t) = Z_{ini}\ (t \leq t_0) \qquad \text{E7.2a}$$

$$Z_r(t) = Z_{ini} + (Z_{max} - Z_{ini})\left(\frac{t - t_0}{t_{max} - t_0}\right)^{1/f_{shape,r}} \qquad (t_0 < t < t_{max}) \qquad \text{E7.2b}$$

$$Z_{ini} = Z_{min}\,\frac{Pct_{Zmin}}{100}, \qquad t_0 = \text{round}(\text{Emergence}/2), \qquad t_{max} = \text{MaxRooting} \qquad \text{E7.3}$$

GDD worked check (T_{max} 34, T_{min} 18, T_{base} 0, T_{upp} 26): method 1 (mean clipped) gives 26 °C·d; methods 2/3 (cap both) give 22 °C·d — the crop file therefore selects the method. Actual root deepening is additionally limited by soil dryness and restrictive layers. Calendar types: days (type 1) or GDD (type 2); stress delays shift t_{adj}. Rainfed sowing rule: plant on the first day in the sowing window where cumulative rain over the previous N days ≥ R and topsoil $\theta \geq \theta_{wp} + 0.5(\theta_{fc} - \theta_{wp})$; otherwise force planting at window end and flag late sowing. Killing frost ($T_{min} < T_{lethal}$) sets status DEAD.

7.3 Canopy cover

VERIFIED IN SOURCE. Growth, saturating and decline forms read from the reference canopy module.

$$CC_0 = \text{PlantPop}\,[\text{plants/ha}] \times \text{SeedSize}\,[\text{cm}^2/\text{plant}] \times 10^{-8} \quad (\text{wheat dump}:\ 4.5\times10^6 \times 1.5 \times 10^{-8} = 0.0675) \qquad \text{E7.4}$$

$$CC(t) = CC_0\, e^{\,CGC\,t} \qquad (CC \leq CC_x/2) \qquad \text{E7.5}$$

$$CC(t) = CC_x - 0.25\,\frac{CC_x^2}{CC_0}\,e^{-CGC\,t} \qquad (CC > CC_x/2) \qquad \text{E7.6}$$

$$CC(t) = CC_x\left[1 - 0.05\left(\exp\left(t_s\,CDC\,\frac{3.33\,(CC_x + 2.29)}{(CC_{x0} + 2.29)\,(CC_x + 2.29)}\right) - 1\right)\right] \quad (t_s = \text{time since senescence start}) \qquad \text{E7.7}$$

t is time since emergence, adjusted for stress delays. Daily increments are the difference of the curve at t and t−1; before senescence CC never decreases. Water stress: canopy expansion uses $CGC_{act} = CGC \cdot K_{s,exp}$ and delays the canopy clock by $(1 - K_{s,exp})$ day (or GDD); the stress-reduced ceiling $CC_{x,W}$ is stored. Early senescence ($K_{s,sen} < 1$) steepens decline via a CDC multiplier calibrated so the reference

daily CC series is reproduced within 0.01.

WARNING. The reference library stores −9.0 in many fields (stage lengths, CGC, CDC, Emergence, Maturity…) meaning “derive at run time”. Code must treat −9 as None, never as a number.

7.4 Water stress coefficients

$$D_{rel} = \mathrm{clip}\Big(\frac{Dr/TAW - p_{up}}{p_{lo} - p_{up}},\ 0,\ 1\Big), \qquad K_s = 1 - \frac{e^{f_{shape} D_{rel}} - 1}{e^{f_{shape}} - 1} \quad (f_{shape} = 0:\ K_s = 1 - D_{rel}) \tag{E7.8}$$

$$p_{adj} = p_{ref} + 0.04\,(5 - ET_0) \ \text{ bounded to } [0, 1] \qquad (\text{flagged, validated on golden data}) \tag{E7.9}$$

Four water stresses with separate (p_{up}, p_{lo}, f_{shape}): canopy expansion ($K_{s,exp}$), stomatal closure ($K_{s,sto}$, drives transpiration), senescence ($K_{s,sen}$) and pollination ($K_{s,pol}$). Aeration stress applies when θ stays above θ_s − Aer/100 for more than 3 days. Germination requires topsoil $\theta \geq \theta_{wp}$ + GermThr·(θ_{fc}−θ_{wp}) (default GermThr 0.2); beyond a maximum delay the crop fails. Heat/cold pollination failure rises logistically between $T_{max,up}$ (≈ 40 °C) and $T_{max,lo}$ (≈ 45 °C), and similarly for cold between $T_{min,up}$ (≈ 10 °C) and $T_{min,lo}$ (≈ 5 °C) — constants live in the crop file and must be confirmed against the reference manual.

7.5 Transpiration and biomass

$$CC^* = \min(1,\ 1.72\,CC - CC^2 + 0.3\,CC^3), \qquad Tr_{pot} = K_{cb}\,CC^*\,ET_0, \qquad Tr_{act} = Tr_{pot}\,K_{s,sto}\,(\times K_{s,aer}) \tag{E7.10}$$

$$dB = WP_{adj}\ \frac{Tr}{ET_0}\ f_{CO_2}\ K_{s,temp}\ K_{s,sal}\ K_{s,N}\ K_{s,PK} \tag{E7.11}$$

$$WP_{adj} = WP(1 - (1 - WP_y/100)\,f_{switch}) \quad (\text{fruit/grain types, during yield formation};\ f_{switch} \in [0,1]) \tag{E7.12}$$

Root extraction distributes Tr_{act} over layers with a depth-weighted sink (maximum S_{top} ≈ 0.048 and S_{bot} ≈ 0.012 m³/m³/day, interpolated), limited by water above θ_{wp}, with unmet demand redistributed to wetter layers. A no-stress shadow run $B_{nostress}$ is carried in parallel for stress reports and the HI pre-flowering factor. Water productivity WP is stored normalised (g/m² per mm of Tr/ET_0); conversion to t/ha is done once at the boundary (10 g/m² = 0.1 t/ha).

DESIGN DECISION. Salinity and fertility stresses multiply the biomass *increment* (conservative; one pathway only, to avoid double-counting with transpiration). The multiplicative combination is a simplification; every factor is logged daily so an agent can explain what limited growth. The Level-3 adapter provides an alternative structure.

7.6 Harvest index and yield

$$HI_{ref}(t) = \frac{HI_{ini}\,HI_0}{HI_{ini} + (HI_0 - HI_{ini})\,e^{-HIGC\,t}}, \qquad HI_{ini} = 0.01, \quad t = \text{days since yield formation start} \tag{E7.13}$$

$$HIGC = \min\left\{h\ (\text{step } 0.001):\ HI_{ref}(t_{YldForm}) > 0.98\,HI_0\right\} \ \Rightarrow\ HI_{ref}(t_{YldForm}) \approx 0.98\,HI_0 \tag{E7.14}$$

$$HI_{adj} = HI_{ref}\,f_{pre}\,f_{post}\,f_{pol} \ \ (\leq HI_0(1 + dHI_0/100)), \qquad Y = B \cdot HI_{adj}, \qquad Y_{fresh} = \frac{Y_{dry}}{1 - f_{moisture}} \tag{E7.15}$$

For root/tuber crops the logistic switches to a linear tail: the switch time is the first t_i where the linear extrapolation HI(t_i) + ($t_{YldForm}$−t_i)(HI(t_i) − HI(t_i−1)) exceeds HI_0, and dHI_{linear} = (HI_0 − HI(t_{switch}))/($t_{YldForm}$ − t_{switch}). f_{pre} encodes pre-flowering stress (moderate canopy-growth stress can raise HI for some crops via a_{HI}; stomatal stress lowers it via b_{HI}); f_{post} encodes yield-formation stress; f_{pol} is the fraction pollinated. Both dry and fresh yields are always reported and labelled.

7.7 Crop library and placeholder policy

Parameters carry value, unit, source {AQUACROP_LIBRARY, FAO_PUBLICATION, LOCAL_TRIAL, EXPERT_ELICITED, PLACEHOLDER}, confidence and range. The hierarchy merge precedence is PLOT > FARM > REGIONAL > GENERIC.

Item	Status
Reference library crops (verified list)	Alfalfa, Barley, Cassava, Cotton, DryBean, Maize, PaddyRice, Potato, Quinoa, Sorghum, Soybean, SugarBeet, SugarCane, Sunflower, Tef, Tomato, Wheat (+ GDD/regional variants)
NOT in the library	pearl millet, mustard, chickpea, cluster bean, cumin, groundnut, sesame, castor, moth/mung bean
Policy for missing crops	analog-crop placeholder (e.g. pearl millet ← sorghum/maize C4 analog) with wide uncertainty, validated=False; recommendations are refused/caveated until calibrated
Level-3 cross-check path	a dedicated pearl-millet/sorghum/wheat module exists in the detailed process model's source tree (structure verified; equations not read)

7.8 Detailed-model adapter (Level 3)

A subprocess adapter writes weather, soil and experiment input files, runs the external executable, and parses summary and daily outputs into the same frame schema. Yield differences > 25 % raise a Validation-agent flag. It is optional and isolated in a container image; input-file formats must be learned from the model's own documentation at implementation time.

7.9 Golden regression values (generated by running the reference model)

VERIFIED IN SOURCE. Reference run: AquaCrop-OSPy 3.1.0, bundled Tunis climate, SandyLoam, Wheat planted 15 Oct, 1979/10/01–1985/05/30, initial water at field capacity.

Season	1	2	3	4	5	6
Rainfed dry yield (t/ha)	8.930	8.393	9.139	8.776	8.710	8.913
Irrigated (SMT 70 %) seasonal irrigation (mm)	36.0	85.9	62.2	111.0	50.0	36.9

Acceptance: yield ±3 %, irrigation ±5 mm, daily canopy cover ±0.01. Failing these blocks promotion of the crop module. The Codex step "extract golden fixtures" also dumps derived values (CC_0, CGC, CDC, stage dates, HIGC, tLinSwitch, dHI_{linear}, daily CC/K_s/biomass) so the derivation code has numeric targets.

8. Salinity and sodicity (original module)

WARNING. The reference crop engine has no salinity model (verified: no salinity code or documentation). This module is therefore an original Level-1 empirical design and is "unvalidated locally" until the user supplies repeated EC_e measurements.

8.1 Units and initial condition

$$TDS\,[\mathrm{mg/L}] \approx 640 \cdot EC \;(EC \le 5\ \mathrm{dS/m}), \qquad 800 \cdot EC \;(EC > 5) \tag{E8.1}$$

$$\mathrm{salt}\,[\mathrm{kg/ha}] = TDS\,[\mathrm{mg/L}] \times h\,[\mathrm{mm}] \times 0.01 \qquad (1\ \mathrm{mm\ over\ 1\ ha} = 10\ \mathrm{m}^3 = 10^4\ \mathrm{L}) \tag{E8.2}$$

$$S_{i,0} = 640\, EC_e\, (1000\, \theta_{s,i}\, \Delta z_i)\, 0.01 = 6400\, EC_e\, \theta_{s,i}\, \Delta z_i \quad [\mathrm{kg/ha}] \tag{E8.3}$$

If only EC(1:5) is known, EC_e = f · $EC_{1:5}$ with a texture-dependent factor from an editable table (ESTIMATED, ±30 %). Worked: EC_e 4 dS/m, θ_s 0.45, Δz 0.1 m → 1152 kg/ha in that 10-cm layer.

8.2 Daily salt step (after the water balance)

- **Input**: irrigation salt = 640·EC_{iw}·gross irrigation (mm)·0.01 kg/ha added to the top layer (rain ≈ 0–20 mg/L).
- **Advection** (complete mixing, top to bottom): with layer water W_i (mm), incoming flux F_{in} and outgoing flux F_i, salt leaving = $S_i \cdot F_i/(W_i + F_{in,i})$. This guarantees S ≥ 0 and is the *same routine* used for nitrate.
- **Evaporation/transpiration** remove water but not salt, so concentration rises automatically.
- **Capillary rise** (if a water table exists) adds 640·EC_{gw}·CR·0.01 kg/ha to the receiving layer.
- Precipitation/dissolution of gypsum and $CaCO_3$, ion pairing and exchange equilibria are **not** modelled at Level 1 (EC > ~8 dS/m with high sulphate over-predicts solubility).

$$EC_{sw,i} = \frac{S_i}{6.4\, W_i}\ \ [\mathrm{dS/m}], \qquad EC_{e,i} = EC_{sw,i}\,\frac{\theta_i}{\theta_{s,i}}, \qquad EC_{e,root} = \sum_i \omega_i\, EC_{e,i}\ \ (\omega = \text{root or } 40/30/20/10 \text{ weights}) \tag{E8.4}$$

8.3 Crop response and sodicity

$$Y_r = 100 - b\,(EC_e - a)\ \ (EC_e > a), \quad Y_r = 100\ (EC_e \le a); \qquad K_{s,sal} = \max(0,\ Y_r/100) \tag{E8.5}$$

$$Y_r = \frac{1}{1 + (EC_e/EC_{50})^p} \qquad (\text{sigmoid variant when parameters are sourced}) \tag{E8.6}$$

$$SAR = \frac{Na}{\sqrt{(Ca + Mg)/2}}\ \ (\mathrm{meq/L}), \qquad ESP = 100\,\frac{-0.0126 + 0.01475\, SAR}{1 + (-0.0126 + 0.01475\, SAR)} \tag{E8.7}$$

$$f_{sodic} = \max(0.1,\ 1 - 0.015\max(0, ESP - 5)) \ \text{ applied to } K_{sat} \text{ when } EC_{iw} < 0.7\ \mathrm{dS/m}\ (\text{flagged empirical}) \tag{E8.8}$$

LITERATURE / UNVERIFIED HERE. Maas–Hoffman threshold–slope parameters (a, b) ship as an editable CSV and are **never invented**. Candidate seeds to be checked against FAO Irrigation & Drainage Paper 29: wheat a = 6.0, b = 7.1; barley a = 8.0, b = 5.0; cotton a = 7.7, b = 5.2. Crops without a value raise MissingDataError. ESP–SAR relation is the USDA Handbook 60 empirical form.

Worked: wheat at EC_e = 10 dS/m → Y_r = 100 − 7.1(10−6) = **71.6 %** (if the seed values are confirmed). Water with Na 10, Ca 3, Mg 2 meq/L → SAR = 6.32, ESP ≈ **7.5 %**.

8.4 Leaching requirement and reclamation

$$LR = \frac{EC_{iw}}{5\, EC_e^{*} - EC_{iw}}, \qquad D_{applied} = \frac{ET}{1 - LR}, \qquad EC_{dw}^{ss} = \frac{EC_{iw}}{LF} \tag{E8.9}$$

Worked: EC_{iw} = 1.5 dS/m, target EC_e = 3 dS/m → LR = 0.111; for ET = 600 mm the applied depth is 675 mm. Steady state with leaching fraction LF = 0.2 gives drainage-water EC = 1.5/0.2 = 7.5 dS/m (a unit test, tolerance 5 %). Reclamation is simulated with the same advection routine under ponded leaching;

an empirical remaining-fraction curve (Reeve–Bower type) has placeholder coefficients that **must** be calibrated locally. Salt mass-balance closure: input − output − ΔS = 0 ± 1 kg/ha.

9. Soil fertility: nitrogen, phosphorus, potassium

WARNING. Also absent from the reference crop engine (verified). Level 1 is mechanistic-simple for N and empirical for P/K; all rate constants below are **priors to be calibrated**, not facts.

9.1 Nitrogen pools and fluxes (per layer, daily)

$$\frac{dU}{dt} = -k_{hyd}\, f_T\, U, \quad k_{hyd} \approx 0.4\ d^{-1}\ (25\,^{\circ}\mathrm{C}), \qquad f_T = 2^{(T-25)/10} \tag{E9.1}$$

$$V_{NH_3} = f_{vol}\, U_{surface} \quad (f_{vol} \approx 0.15 \pm 0.10;\ \uparrow \text{with pH} > 7.5;\ = 0 \text{ if incorporated or irrigated within } 24\,\text{h}) \tag{E9.2}$$

$$\frac{d\,NH_4}{dt}\,(\text{nitrification}) = -k_{nit}\, f_T\, f_\theta\, NH_4, \quad k_{nit} \approx 0.2\ d^{-1}; \quad f_\theta:\ 0\ (\theta \le \theta_{wp}) \nearrow 1\ (\approx 0.6\,\theta_s) \searrow 0.5\ (\theta_s) \tag{E9.3}$$

$$N_{min} = \sum_{pools} \frac{C_{decomp,\,pool}}{CN_{pool}} - \frac{C_{\to BIO} + C_{\to HUM}}{CN_{BH}}, \qquad CN_{BIO} = CN_{HUM} = 8.5 \tag{E9.4}$$

$$NO_3^{leach,\,i} = NO_{3,\,i}\, \frac{F_i}{W_i + F_{in,\,i}}, \qquad \text{denitrification} = k_{den}\, NO_3 \quad (\theta > 0.9\theta_s;\ k_{den} \approx 0.02\ d^{-1}) \tag{E9.5}$$

Net N_{min} < 0 is immobilisation (limited by available mineral N). Residue C:N inputs: cereal straw ≈ 80, legume residue ≈ 20, FYM ≈ 15 (priors). Legume fixation N_{fix} = fraction$_{Ndfa}$ × crop N uptake (crop priors, flagged). Atmospheric deposition prior ≈ 5 kg N/ha/yr. Nitrate leaving the root zone is a loss reported in kg N/ha.

9.2 Crop demand, uptake and stress

$$N_c(W) = a\, W^{-b} \quad (W \text{ in t DM/ha};\ \text{wheat } a = 5.35\%,\ b = 0.442), \qquad N_{demand} = N_c(W)\, dW \tag{E9.6}$$

$$U_{cap,\,i} = (NH_{4,\,i} + NO_{3,\,i})\, \phi_i\, f_{uptake}, \qquad U_{act} = \min(N_{demand}, \sum_i U_{cap,\,i}), \qquad K_{s,\,N} = \mathrm{clip}(\frac{\sum U_{act}}{\sum N_{demand}}, 0, 1) \tag{E9.7}$$

Worked (wheat, literature coefficients): W = 5 t/ha → $N_c = 5.35 \cdot 5^{-0.442}$ = 2.63 %, demand = **131 kg N/ha**. Effect: dB × $K_{s,N}$, optionally CGC × $K_{s,N}^{½}$. Defaults for crops with no coefficients (a = 4.5, b = 0.4) are flagged *placeholder* and block production use until set.

LITERATURE / UNVERIFIED HERE. Justes-type dilution coefficients, k_{hyd}, k_{nit}, k_{den}, f_{vol} and C:N values are priors written from domain knowledge; confirm against sources and calibrate.

9.3 Phosphorus, potassium, micronutrients, recommendations

$$RY = 1 - \exp[-c\,(S_{index} + e\,F)] \qquad (\text{crop specific } c,\ \text{fertiliser efficiency } e;\ \text{multiplier on biomass}) \tag{E9.8}$$

- Soil-test method compatibility is checked before use: Olsen-P and Bray-P thresholds differ; alkaline-permanganate N is an *index*, not mineral N.
- Critical-level and class tables are editable CSVs approved by an agronomist — **no hard-coded state advisories**.
- Micronutrients (Zn, Fe, B, S…) are flag-only at Level 1: soil test vs critical level → deficiency_risk indicator and amendment line items, no yield simulation.
- Recommendation logic is model-in-the-loop: simulate N rates 0…R_{max} in 20 kg steps under weather scenarios and choose the economic optimum (marginal revenue = marginal cost), with a CVaR variant. Prior efficiency 0.4–0.6, higher with split applications.

9.4 Amendments and organic inputs

$$GR\,[\mathrm{cmol_c/kg}] = CEC\,\frac{ESP_i - ESP_t}{100}, \qquad M_{gyp}\,[\mathrm{kg/ha}] = \frac{GR \cdot 8610 \cdot d\,[\mathrm{m}] \cdot \rho_b\,[\mathrm{t/m^3}]}{\text{purity} \cdot \text{efficiency}} \tag{E9.9}$$

Derivation: 1 $cmol_c$/kg of exchangeable Na replaced needs 0.5 cmol of $CaSO_4 \cdot 2H_2O$ (172.17 g/mol) = 0.861 g per kg soil; soil mass per hectare = $10^4 \cdot d \cdot \rho_b \cdot 1000$ kg, giving the 8610 constant. **Worked:** GR = 1, d = 0.15 m, ρ_b = 1.4 → **1808 kg/ha** at purity = efficiency = 1 (default efficiency 0.75 is flagged). A CSV-driven decision table picks the path: sodic (ESP > 15, EC < 4, pH > 8.2) → gypsum + leaching; saline-sodic → gypsum then leaching; saline (EC > 4, ESP < 15) → leaching only; acid (pH < 5.5) → lime. Dose bounds: gypsum ≤ 25 t/ha/yr, FYM ≤ 30 t/ha, N ≤ crop maximum. Organic inputs feed RothC (carbon), the nitrogen module (organic N) and water properties (OM% updated annually → pedotransfer recomputed).

10. Soil organic carbon (RothC-type, monthly)

VERIFIED IN SOURCE. Rate constants, temperature factor, TSMD limit and partitioning were read from the 342-line pyRothC implementation (CC0). The vegetation-cover factor was **not** applied there — see the decision below.

Five pools in the modelled topsoil (default 23 cm): DPM, RPM, BIO, HUM (all t C/ha) and inert IOM (k = 0).

$$k_{DPM} = 10, \quad k_{RPM} = 0.3, \quad k_{BIO} = 0.66, \quad k_{HUM} = 0.02, \quad k_{IOM} = 0 \quad [\mathrm{yr}^{-1}] \tag{E10.1}$$

$$a = \frac{47.91}{1 + \exp(106.06/(T + 18.27))} \quad (a = 0 \text{ if } T < -5\,^{\circ}\mathrm{C}), \qquad C(t + \tfrac{1}{12}) = C(t)\exp(-k\,a\,b\,c/12) \tag{E10.2}$$

$$TSMD_{max} = -(20 + 1.3\,\mathrm{clay} - 0.01\,\mathrm{clay}^2)\,\frac{d_{cm}}{23} \quad (\text{bare soil}: \ \div 1.8) \tag{E10.3}$$

$$b = 1 \ \ (TSMD > 0.444\,TSMD_{max}), \qquad b = 0.2 + 0.8\,\frac{TSMD_{max} - TSMD}{TSMD_{max} - 0.444\,TSMD_{max}} \ \ (\text{otherwise; floor } 0.2) \tag{E10.4}$$

$$x = 1.67\,(1.85 + 1.60\,e^{\,-0.0786\,\mathrm{clay}}), \qquad f_{CO_2} = \frac{x}{x+1}, \quad f_{BIO} = \frac{0.46}{x+1}, \quad f_{HUM} = \frac{0.54}{x+1} \tag{E10.5}$$

Each of DPM, RPM, BIO and HUM decomposes into CO_2, BIO and HUM with fractions E10.5; BIO and HUM products re-enter their own pools. Moisture driver: the preferred input is actual root-zone deficit from the daily water model aggregated monthly; the classical TSMD form (rain − 0.75 × open-water evaporation, bounded below by $TSMD_{max}$) is the fallback.

Worked value	Result
clay = 20 %: x, CO_2 / BIO / HUM fractions	x = 3.644; 0.785 / 0.099 / 0.116
rate factor a at 10 / 25 / 35 °C	1.10 / 3.80 / 5.76
$TSMD_{max}$ (clay 20 %, 23 cm)	-42 mm
TOC = OC% · BD · depth_cm for OC 0.5 %, BD 1.45, 23 cm	16.7 t C/ha
IOM = $0.049 \cdot TOC^{1.139}$ (literature)	1.21 t C/ha

10.1 Inputs, initialisation, outputs

$$C_{in} = 0.45 \times \text{residue DM (t/ha)} \ \ (\text{measured C fraction overrides}), \qquad DPM:RPM = 1.44 \ (\text{crops}), \ 0.25 \ (\text{woodland}) \tag{E10.6}$$

$$FYM: \ DPM:RPM:HUM = 0.49:0.49:0.02, \qquad TOC = OC\% \cdot \rho_b \cdot d_{cm} \ \ [\mathrm{t\,C/ha}], \qquad IOM = 0.049\,TOC^{1.139} \tag{E10.7}$$

- **Initialisation**: only TOC is usually measured, so the default is an equilibrium spin-up: bisect on C_{in} so the model reaches TOC under long-term mean climate with IOM fixed. Pool split is therefore ESTIMATED (wide) even though TOC is MEASURED.
- **Root biomass** is not simulated by the crop engine, so residue carbon uses crop-specific root:shoot ratios (placeholders).
- **Outputs**: monthly pools, CO_2-C respired, ΔSOC at 1/5/10 yr with Monte-Carlo uncertainty (C_{in} and k multipliers lognormal CV 20 %, clay ± 3 %); carbon reported as tC and $tCO_2e = 3.667 \cdot tC$ — never as credits.
- **Tests**: C mass balance $C_{in} - CO_2 = \Delta C$ within 1e-9; one-pool analytic $C(t) = C_0 e^{-10t}$ (a = b = c = 1); equilibrium after 1000 years within 0.1 %.

DESIGN DECISION. Original RothC documentation applies a plant-cover factor c = 0.6 (vegetated) / 1.0 (bare); the Python port inspected does not. This project implements c as a configurable switch, **default ON**, with unit tests for both modes and a note in docs/model_decisions.md. Under agrivoltaic shade, the soil-temperature offset and reduced-ET moisture flow into a and b.

11. Solar PV energy model (hourly)

An original implementation of the standard chain; the pvlib package is an *oracle in tests only* (optional dependency), not a runtime dependency. Golden clear-sky values were generated with it (Section 11.8).

11.1 Solar position and extraterrestrial radiation

Michalsky/Meeus-grade algorithm (error < 0.01°): Julian day → mean longitude and anomaly → ecliptic longitude → obliquity → right ascension and declination → equation of time → hour angle → zenith and azimuth (from north, clockwise) → refraction correction. Validation at Jodhpur (26.2389° N, 73.0243° E): solar-noon zenith 5.39° on 2026-06-21, 50.00° on 2026-12-21, 26.20° on 2026-03-21 (tolerance 0.1°).

$$G_0 = 1367\,(1 + 0.033\cos(2\pi\,doy/365)) \quad [\mathrm{W/m^2}] \tag{E11.1}$$

11.2 Irradiance decomposition (Erbs)

$$k_t = \frac{GHI}{G_0 \cos\theta_z}, \qquad \frac{DHI}{GHI} = 1 - 0.09\,k_t \quad (k_t \le 0.22) \tag{E11.2}$$

$$\frac{DHI}{GHI} = 0.9511 - 0.1604\,k_t + 4.388\,k_t^2 - 16.638\,k_t^3 + 12.336\,k_t^4 \quad (0.22 < k_t \le 0.80), \qquad 0.165 \quad (k_t > 0.80) \tag{E11.3}$$

$$DNI = \frac{GHI - DHI}{\cos\theta_z} \tag{E11.4}$$

Clear-sky planning model: Ineichen–Perez, with Haurwitz GHI = 1098 cos θ_z exp(−0.059/cos θ_z) as a simple fallback. **Rajasthan aerosol load is high, so clear-sky models over-estimate**; use measured or reanalysis GHI wherever possible.

11.3 Transposition to the plane of array

$$\cos AOI = \cos\theta_z\cos\beta + \sin\theta_z\sin\beta\cos(\gamma_s - \gamma_p), \qquad E_b = DNI \cdot \max(0, \cos AOI) \tag{E11.5}$$

$$\epsilon = \frac{(DHI + DNI)/DHI + 1.041\,\theta_z^3}{1 + 1.041\,\theta_z^3} \quad (\theta_z \text{ in rad}), \qquad F_1 = \max(0, F_{11} + F_{12}\Delta + F_{13}\theta_z), \quad F_2 = F_{21} + F_{22}\Delta + F_{23}\theta_z \tag{E11.6}$$

$$E_d = DHI\Big[(1 - F_1)\frac{1 + \cos\beta}{2} + F_1\,\frac{a}{b} + F_2\sin\beta\Big], \quad a = \max(0, \cos AOI), \quad b = \max(\cos 85^\circ, \cos\theta_z) \tag{E11.7}$$

$$E_g = GHI \cdot \rho\,\frac{1 - \cos\beta}{2} \quad (\rho = \text{albedo},\ 0.25 \text{ dry soil}), \qquad POA = E_b\,IAM_b + E_d\,IAM_d + E_g\,IAM_g \tag{E11.8}$$

LITERATURE / UNVERIFIED HERE. The Perez (1990) coefficient table F_{11}...F_{23} by ε-bin (8 bins) must be typed from the paper; Codex adds a 24-point comparison with an independent implementation to 1 W/m². Isotropic fallback: E_d = DHI(1+cos β)/2. Brightness Δ is the usual air-mass–scaled sky brightness.

$$IAM_b = 1 - b_0\left(\frac{1}{\cos AOI} - 1\right), \ b_0 = 0.05 \text{ (ASHRAE)}; \qquad \theta_{eff,\,sky} = 59.7 - 0.1388\beta + 0.001497\beta^2, \quad \theta_{eff,\,gnd} = 90 - 0.5788\beta + 0.002693\beta^2 \tag{E11.9}$$

11.4 Soiling, cell temperature, DC and AC power

$$\text{Faiman}: \quad T_{cell} = T_{amb} + \frac{G}{U_0 + U_1\,v}, \quad U_0 = 25\ \mathrm{W/m^2K},\ U_1 = 6.84\ \mathrm{W\,s/m^3K} \tag{E11.10}$$

$$\text{SAPM}: \quad T_m = G\,e^{a + b\,v} + T_{amb}, \quad T_{cell} = T_m + \frac{G}{1000}\Delta T \ \ (\text{open rack glass/glass}: \ a = -3.47,\ b = -0.0594,\ \Delta T = 3) \tag{E11.11}$$

$$P_{dc} = \frac{G_{eff}}{1000}\,P_{dc0}\,[1 + \gamma\,(T_{cell} - 25)], \quad \gamma \approx -0.004\ \mathrm{K^{-1}}, \qquad G_{eff} = G_{front} + \varphi\,G_{rear}\,(1 - l_{rear}) \tag{E11.12}$$

$$\zeta = \frac{P_{dc}}{P_{dc0,\,inv}}, \qquad \eta = \frac{\eta_{nom}}{\eta_{ref}}\Big(-0.0162\,\zeta - \frac{0.0059}{\zeta} + 0.9858\Big), \quad \eta_{ref} = 0.9637, \qquad P_{ac} = \min(\eta\,P_{dc},\ P_{ac0}) \tag{E11.13}$$

$$P(t) = P_0\,(1-d)^t, \quad d \approx 0.005\ yr^{-1}; \qquad PR = \frac{E_{ac}}{P_{dc0}\,[\mathrm{kW}] \cdot H_{POA}\,[\mathrm{kWh/m^2}]} \tag{E11.14}$$

Soiling: loss accumulates with dry days (prior 0.2–0.5 %/day in the Rajasthan dust belt, calibratable), reset largely by rain ≥ 5 mm or cleaning events (cost to finance). Other multiplicative losses (flagged defaults): mismatch 2 %, DC wiring 2 %, connections 0.5 %, nameplate 1 %, light-induced degradation 1.5 % in year 1, availability 3 %. Bifacial rear irradiance comes from view factors in the agrivoltaic module. Uncertainty: interannual (weather scenarios), model (±3–5 %), soiling (distribution), degradation (triangular 0.3–0.8 %/yr).

11.5 Layout and sizing for tiny plots

$$w = W_c \cos\beta, \qquad \Delta h = W_c \sin\beta, \qquad GCR = \frac{W_c}{P} \tag{E11.15}$$

$$P_{min} = W_c\left(\cos\beta + \frac{\sin\beta}{\tan\alpha_s}\right) \qquad (\alpha_s = \text{limiting sun elevation computed, not a rule of thumb}) \tag{E11.16}$$

Worked: β = 20°, W_c = 2 m, α_s = 25° → P_{min} = 2(0.9397 + 0.3420/0.4663) = **3.35 m**. Rows are fitted inside the setback-buffered polygon along the long axis; modules per row = ⌊row length / module width⌋. Row-to-row shading reduces beam on affected strings in bypass-diode steps. Electrical checks (DC/AC ratio 1.1–1.3, string voltage within the MPPT window using V_{oc} temperature correction) are *checks only* — a licensed installer must confirm. Single-axis tracking with backtracking is an optional configuration, not the default for tiny plots.

11.6 Golden clear-sky values (Jodhpur, 1 kWp)

VERIFIED IN SOURCE. Generated by running pvlib 0.16.1: tilt 26° south, albedo 0.25, Ineichen clear sky, Perez, SAPM open-rack glass/glass, T_{amb} 30 °C, wind 2 m/s, γ = −0.004, PVWatts inverter P_{ac0} = 1000 W.

Date	GHI kWh/m²	POA kWh/m²	DC kWh	AC kWh	Noon zenith
2026-03-21	6.578	7.400	6.515	6.254	26.20°
2026-06-21	7.669	6.828	6.097	5.848	5.39°
2026-12-21	4.054	6.078	5.410	5.198	50.00°

Tolerances: GHI/POA ±2 %, DC/AC ±3 % (clear-sky, transposition and turbidity choices differ between implementations). Note how winter POA (6.08) exceeds winter GHI (4.05) — the tilt is doing its job.

12. Agrivoltaics: ground radiation and crop coupling

WARNING. No open-source crop model supplying radiation beneath PV rows was found among the projects inspected (agrivoltaic repositories were listed but their code was not read). The geometry below is an original 2-D design checked by an independent 3-D ray-casting oracle. The **crop response to shade is EXPERIMENTAL and unvalidated**; every result carries the flag `agrivoltaic_unvalidated` until local trial data exist.

12.1 Geometry (infinite rows, cross-section plane)

Each row is a strip of slant width W_c, tilt β, azimuth γ_p, lower edge at clearance h_0, upper edge at $h_0 + W_c \sin\beta$, repeating every pitch P. Ground samples x_k are placed every 0.10 m across one pitch (capped at 400 points).

$$\mathbf{s} = (\cos\alpha\sin\gamma_s,\ \cos\alpha\cos\gamma_s,\ \sin\alpha), \qquad \phi = \operatorname{atan2}(s_\perp,\ s_z)\ \ \text{(profile angle in the plane } \perp \text{ to the row axis)} \tag{E12.1}$$

$$x'_{edge} = x_{edge} - z_{edge}\tan\phi, \qquad \text{shaded interval} = [\min, \max] \text{ of projected edges } (+ \text{ periodic copies at } \pm P,\ \pm 2P,\ \dots) \tag{E12.2}$$

$$\tau_{eff} = g + (1-g)\,\tau_{cell} \qquad (g = \text{module gap fraction};\ \text{opaque panel}:\ \tau_{eff} = 0) \tag{E12.3}$$

12.2 Diffuse sky and ground irradiance

$$SVF(x_k) = 1 - \sum_{strips} \frac{\sin\theta_2 - \sin\theta_1}{2} \tag{E12.4}$$

$$G_g(x_k, t) = DNI\cos\theta_z\,\tau_{beam}(x_k, t) + DHI \cdot SVF(x_k) \quad (+ \text{ transmitted terms}) \tag{E12.5}$$

$$PAR = f_{par}\,G_g \quad (f_{par} \approx 0.45;\ \text{range } 0.43\text{–}0.50), \qquad F_{rad}(x, d) = \frac{\sum_h G_g}{\sum_h GHI_{open}} \tag{E12.6}$$

θ_1, θ_2 are the angles (from the vertical, in the cross-row plane) subtended by each strip's edges as seen from the ground point. The 2-D result is exact for a horizontal receiver and infinite parallel strips. Level 1 uses an isotropic sky; anisotropic sky and finite-row end effects are Level 2. Panel-underside emission and ground–ground reflection are neglected; rows shorter than ~4h_0 are flagged "upper bound on shading". Daily F_{rad} is averaged over each crop strip (area-weighted); only the strip series plus full profiles for five representative days per season are stored.

Analytic checks used as tests

$$\text{overhead sun, opaque horizontal panel}:\ \ \overline{\tau_{beam}} = 1 - \frac{w}{P} \tag{E12.7}$$

$$\text{point beneath centre of a horizontal strip of width } 2a \text{ at height } h:\ \ SVF = 1 - \frac{a}{\sqrt{a^2 + h^2}} \tag{E12.8}$$

Also: no panels → F_{rad} = 1; a brute-force 3-D ray-casting oracle must agree with the 2-D profile-angle result within 1 % of shaded area.

12.3 Crop coupling mechanisms (each switchable)

	Mechanism	Implementation	Status
A	atmospheric demand	ET_0 recomputed with $R_s \cdot F_{rad}$, u_2·wind factor (prior 0.6–1.0), $T_{max} + \Delta T_{max}$ (prior −0.5…−3 °C), $T_{min} + \Delta T_{min}$ (0…+1 °C)	priors flagged
B	light-limited growth	$dB = \min(dB_{water}, RUE \cdot fPAR \cdot PAR_{ground})$; fPAR ≈ CC; RUE C3 ≈ 2.5–3.0, C4 ≈ 3.0–3.8 g DM/MJ PAR (literature ranges)	calibrate
C	shade-adaptive response	$CGC_{shade} = CGC(1 + k_s(1-F_{rad}))$; HI penalty c_h; defaults $k_s = c_h = 0$ (no adaptation) + warning	calibration required

	Mechanism	Implementation	Status
D	soil temperature / moisture	ΔT_{soil} to RothC and emergence; reduced Es via shaded ET_0	priors
E	rain redistribution	drip-line concentration at the lower edge; mean multiplier 1 across the pitch; default OFF	flagged
F	spatial strips	split plot into under-panel / edge / inter-row strips (default 3, max 8), run crop engine per strip, area-weight the yield	structural

$$RY = \frac{Y_{AV}}{Y_{open}}, \qquad LER = RY + \frac{E_{AV}}{E_{ref}}, \qquad LER_{econ} = \frac{R_{AV}}{\max(R_{crop,\,open},\ R_{PV,\,ground})} \tag{E12.9}$$

LER > 1 means joint use beats separate use of equal areas; E_{ref} is the energy of a standard ground array on the same area at its own GCR. Printed with every result: simplified light–water coupling, no crop-specific field-validated shade function, sensitivity to RUE, shade tolerance and k_s, and a requirement to run a trial plot with PAR measurements (ceptometer/lux meter) before capital commitment. User-uploaded PAR readings are MEASURED and are used to fit τ_{eff} and f_{par} by least squares.

13. Finance and economics

Annual-step discounted cashflow with optional monthly resolution for working-capital checks. All inputs are nominal INR with explicit inflation per series (default 5 % general; tariff escalation 0–3 % user-set). **The code ships no hard-coded prices, tariffs or subsidies**: missing economic inputs raise MissingDataError naming exactly what is needed, and every report prints currency, price basis, horizon and discount rate. ACTUAL and SIMULATED cashflow entries are never summed unless a report says so.

13.1 Core metrics

$$NPV = \sum_{t=0}^{T} \frac{CF_t}{(1+r)^t}, \qquad IRR: \quad NPV(IRR) = 0 \ \ (\text{Brent on } [-0.99,\ 10]) \tag{E13.1}$$

$$EAA = NPV \frac{r}{1-(1+r)^{-T}}, \qquad BCR = \frac{PV(\text{benefits})}{PV(\text{costs})} \tag{E13.2}$$

$$LCOE = \frac{CAPEX + \sum_t Opex_t/(1+r)^t}{\sum_t E_t/(1+r)^t} \ \ [\text{INR/kWh}], \qquad EMI = \frac{Pi\,(1+i)^n}{(1+i)^n - 1} \tag{E13.3}$$

$$E_{pump}\,[\text{kWh/m}^3] = \frac{\rho g H}{3.6\times 10^6\,\eta} \approx \frac{0.002725\,H}{\eta}, \qquad CVaR_\alpha = \eta^* - \frac{1}{1-\alpha}\,E[(\eta^* - NPV)^+] \tag{E13.4}$$

Worked check	Result
CF = −1000, then +300 for 5 years, r = 10 %	NPV = 137.24; IRR = 15.24 %
₹1,00,000 loan, 12 % p.a., 12 monthly instalments (i = 0.01)	EMI = ₹8885
pump head 30 m, total efficiency 0.5	0.1635 kWh/m³

13.2 Crop budget, PV project, agrivoltaic combination

- **Crop budget per plot-season**: revenue = yield (t/ha) × area (ha) × farm-gate price (mandi modal − transport − commission; price is a scenario draw; fresh vs dry basis stated). Costs: seed, fertiliser, amendments, pesticide, labour (days × wage), water and pumping (kWh × tariff), machinery, insurance, interest on working capital, lease.
- **PV project**: year-0 capex; yearly revenue = self-consumed kWh × avoided tariff + exported kWh × export tariff under the user-entered tariff rule (net-metering with banking and credit expiry); opex (cleaning, O&M % of capex, insurance); inverter replacement around year 10–12; degradation 0.5 %/yr; subsidies only as explicit user-entered lines.

- **Agrivoltaic**: crop revenue under panels (from the coupling module) + PV revenue; shared structure cost; reduced-mechanisation labour penalty (user parameter).
- **Risk and sensitivity**: NPV distribution over joint weather × price × tariff scenarios; VaR and CVaR at 0.80/0.90; P(NPV < 0); worst-year cash shortfall probability; downside deviation; tornado (±20 % on price, yield, capex, tariff, discount rate) and Sobol indices.

13.3 Water and energy costs

Pumping energy per E13.4 with H = static lift + drawdown + friction/pressure head. A groundwater-depletion shadow price (INR/m³) is optional (default 0). The irrigation planner converts monthly net requirement to gross using system efficiency, compares with the availability series, and evaluates yield penalties by *running the crop engine*, never a linear shortcut.

14. Uncertainty quantification

Three uncertainty sources are kept separate so reports can say *where the risk comes from*: (A) weather — aleatory; (B) parameters and structure — epistemic (posterior draws); (C) economics — prices, tariffs, costs.

14.1 Weather scenario generation

- **Historical replay** (default for season decisions): each historical year is a scenario, preserving inter-variable correlation (N ≈ 35).
- **Block bootstrap** for multi-year horizons: resample years or 15–30-day monthly blocks (N = 200–1000).
- **Parametric generator** when long records are missing (WGEN-type, below). **Climate shift**: delta factors on any base set (monsoon rain scale, +ΔT, heat-wave frequency) — labelled scenarios, never forecasts. **Seasonal-forecast conditioning** (experimental): analog reweighting bounded to [0.5, 2].

$$P(\text{wet}_t|\text{dry}_{t-1}) = p_{01}(m), \quad P(\text{wet}_t|\text{wet}_{t-1}) = p_{11}(m), \quad R_t \sim \text{Gamma}(k_m, \theta_m) \ \ (\text{wet days}) \qquad \text{E14.1}$$

$$\varepsilon_t = \rho\,\varepsilon_{t-1} + \sqrt{1-\rho^2}\,\xi_t, \qquad T_{max/min} = \mu_m(\text{state}) + \sigma_m(\text{state})\,\varepsilon_t \qquad \text{E14.2}$$

$$\ln P_t = \mu_{m(t)} + \varphi\,(\ln P_{t-1} - \mu_{m(t-1)}) + \sigma\,\xi_t \qquad (\text{price AR(1) with seasonal mean; fitted on} \geq 5 \text{ years}) \qquad \text{E14.3}$$

Generator diagnostics must match observations: mean annual rain ±10 %, wet-day frequency, dry-spell P90 ±15 %, T_{max}/T_{min} means. The same draws are shared across plots and across candidate plans (**common random numbers**) so scenario differences have low variance.

14.2 Propagation and variance attribution

$$Var(Y) = E_w[Var(Y|\,w)] + Var_w(E[Y|\,w]) \qquad (\text{nested over weather w, parameters } \theta, \text{ prices } \pi) \qquad \text{E14.4}$$

Default budget: 50 weather × 20 parameter draws = 1000 simulations (parameters by Latin hypercube; economics by Sobol sequence, N_{price} larger because post-processing is cheap). N auto-increases until the bootstrap relative SE of the mean and of the 5th percentile is < 2 % (capped). Outputs: mean, sd, quantiles (5, 25, 50, 75, 95), P(Y < threshold), CVaR — and always the samples, for downstream optimisation.

14.3 Global sensitivity analysis

$$S_i = \frac{V - \frac{1}{2}E[(f(B) - f(A_B^i))^2]}{V}, \qquad S_{T,i} = \frac{\frac{1}{2}E[(f(A) - f(A_B^i))^2]}{V} \qquad (\text{Jansen estimators, Saltelli sampling } N(2D+2)) \qquad \text{E14.5}$$

$$\mu_i^* = \frac{1}{r}\sum_{j=1}^{r}|EE_i^{(j)}|, \qquad \sigma_i = \text{sd}(EE_i^{(j)}) \qquad (\text{Morris}, \ r = 20) \qquad \text{E14.6}$$

Validation: the Ishigami function has analytic indices S_1 = (0.314, 0.442, 0) and S_T = (0.558, 0.442, 0.244); the implementation must recover them within 0.03 with N = 2^{14}. Results rank inputs (soil EC,

water EC, rainfall, price, discount rate, τ_{eff}...) and feed value-of-information.

15. Calibration, validation and safe learning

15.1 Four-level hierarchy with partial pooling

$$\theta_j^G \sim N(\mu_g, \sigma_g^2), \quad \theta_j^R \sim N(\theta_j^G, \tau_r^2), \quad \theta_j^F \sim N(\theta_j^R, \tau_f^2), \quad \theta_{j,k}^P \sim N(\theta_j^F, \tau_p^2) \qquad \text{E15.1}$$

$$y_{k,m} \sim N(f_m(\theta_k^P,\, x),\ \sigma_m^2), \qquad \sigma_m^2 = \sigma_{meas}^2 + \sigma_{struct}^2 \quad (\sigma_{struct}\ \text{estimated},\ \text{inverse} - \text{gamma prior}) \qquad \text{E15.2}$$

$$w = \frac{n_k/\sigma^2}{n_k/\sigma^2 + 1/\tau_p^2} \qquad (\text{posterior mean } = w \cdot \text{plot data} + (1-w) \cdot \text{farm value}) \qquad \text{E15.3}$$

Parameters are transformed (log for rates, logit for fractions). Half-normal hyper-priors on τ are scaled so that with no data a plot shrinks ~90 % to the farm value. Levels: GENERIC (published) → REGIONAL (arid NW India) → FARM → PLOT. **Calibratable sets are deliberately small**: crop {CGC, CC_x, WP or RUE, HI_0, Z_{max}, K_{cb}, stomatal p}; soil {K_{sat}/τ, REW, θ_{fc} offset}; carbon {k multipliers}; salinity {a, b}; nutrients {k_{nit}, f_{uptake}, mineralisation multiplier}; PV {soiling rate, γ, system loss}; agrivoltaic {τ_{eff}, f_{par}, wind factor, ΔT}. Rule: plot-level fitting requires $n_{obs} \geq 5 \times n_{params}$. Parameters with posterior contraction < 20 % are reported "not identified by these data" and left at the parent level.

15.2 Inference methods

- **Ensemble MCMC** (affine-invariant, emcee-class) over ≤ 8 parameters of one level, conditional on the parent posterior; ≥ 32 walkers, ≥ 2000 steps; convergence $\hat{R} < 1.05$ and ESS > 400.
- **ABC-SMC** when the likelihood is awkward (e.g. only season yield is observed); distance = normalised RMSE across observed variables.
- **GLUE** only as a diagnostic, never for promotion. Prior-predictive check p < 0.01 flags prior–data conflict for expert review instead of silently widening. Twin experiments: recover known parameters in ≥ 90 % of 100 trials within 95 % intervals.

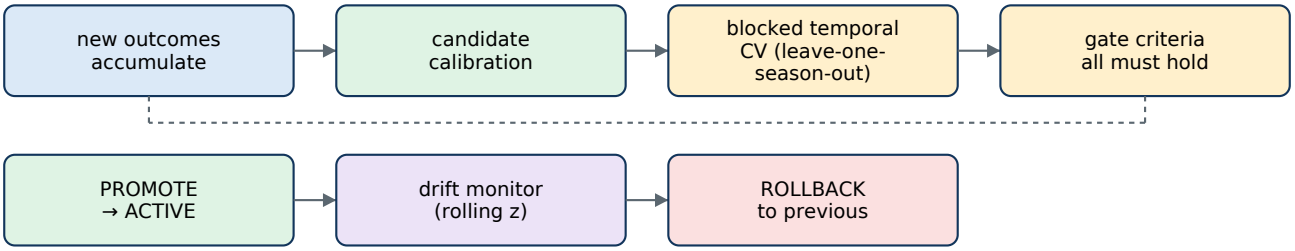
15.3 Scoring

$$NSE = 1 - \frac{\sum(o-p)^2}{\sum(o-\bar{o})^2}, \qquad KGE = 1 - \sqrt{(r-1)^2 + (\alpha-1)^2 + (\beta-1)^2}, \quad \alpha = \frac{s_p}{s_o},\ \beta = \frac{\bar{p}}{\bar{o}} \qquad \text{E15.4}$$

$$CRPS = E|X-y| - \tfrac{1}{2}E|X-X'| \quad (\text{fair ensemble form}), \qquad \text{pinball}_\tau = \max(\tau(y-q),\ (\tau-1)(y-q)) \qquad \text{E15.5}$$

$$\text{conformal interval half} - \text{width} = \hat{q}_{\lceil (n+1)(1-a) \rceil / n} \quad (\text{split conformal on blocked residuals}) \qquad \text{E15.6}$$

15.4 The promotion gate (no candidate is used unless it proves itself)



Criterion (all must hold)	Threshold
pooled held-out CRPS	candidate ≤ 0.95 × parent
per-variable RMSE	no variable worsens by more than 10 %
90 % interval coverage	between 0.80 and 0.97 (PIT check)
physical sanity	parameters in range; mass balances unchanged

Criterion (all must hold)	Threshold
minimum evidence	≥ 2 independent seasons or ≥ 8 independent observation points at that level
stability	change < 3 posterior SD vs the previous promoted set unless justified
drift (after promotion)	|mean z| > 2 for 3 consecutive outcomes, or coverage collapse → automatic rollback

Agents cannot promote parameters; only the gate (or an admin with an approval token) can. Decisions and reasons are stored in calibration_run and reported to the user in plain language ("model changed because…" or "model unchanged because it did not beat the current version").

15.5 Residual machine learning (only after enough data)

- Physics first; the ML model learns the residual $r = y_{obs} - y_{model}$. **Training is refused below 30 independent labelled outcomes** for the target variable.
- Shallow gradient-boosted trees (depth ≤ 4, strong regularisation, monotonic constraints where obvious) or a small Gaussian process; features use only information available at prediction time (no leakage).
- Intervals by split-conformal or quantile regression; the same promotion gate decides use; SHAP summaries stored; **extrapolation guard**: outside the 1st–99th-percentile training envelope the correction is NaN and intervals widen.

15.6 Closing the loop: error attribution

After an outcome arrives, the same plan is re-run with *actual* weather and *actual* interventions (hindcast). Total error then splits into (i) weather realisation, (ii) management deviation and (iii) residual model/parameter error. **Only (iii) feeds calibration.** For each outcome the system also stores z = (obs − mean)/sd, the PIT value and whether the observation fell in the 90 % interval — the "learning ledger" shown in the UI as *Is the twin getting better?*

16. Optimisation and value of information

16.1 Decision structure

Stage 1 (here-and-now): $x_{p,o} \in \{0,1\}$ — plot p adopts option o (crop bundle with irrigation and fertility plan, ground PV, agrivoltaic, reclamation-fallow, water storage, or sub-area splits at e.g. 25 % resolution). Stage 2 (recourse) lives inside the simulator's irrigation policy. The expensive non-linear agronomy is **pre-computed into a payoff table** per (plot, option, scenario): NPV, water by month, capex by year, carbon change, end-of-horizon EC_e, labour. The optimiser is therefore a linear model over x. Cross-plot effects (shade on a neighbour, shared pump) are handled as joint pair options or farm-level constraints.

16.2 Risk-aware MILP (Rockafellar–Uryasev CVaR)

$$\max_{x,\,\eta,\,z} \quad (1-\lambda)\sum_s \pi_s\, NPV_s + \lambda\left(\eta - \frac{1}{1-\alpha}\sum_s \pi_s z_s\right), \qquad NPV_s = \sum_{p,\,o} payoff_{p,\,o,\,s}\, x_{p,\,o} \tag{E16.1}$$

$$z_s \geq \eta - NPV_s, \quad z_s \geq 0; \qquad \sum_o x_{p,\,o} = 1 \;\; \forall p; \qquad \sum_{p,\,o} capex_{p,\,o}\, x_{p,\,o} \leq Budget; \qquad \sum_{p,\,o} water_{p,\,o,\,s,\,m}\, x_{p,\,o} \leq Supply_{s,\,m} + Purchase_{s,\,m} \tag{E16.2}$$

VERIFIED IN SOURCE. This formulation was prototyped and solved with Pyomo 6.10.1 + HiGHS (appsi_highs): 3 plots × 3 options (crop / PV / agrivoltaic) × 50 scenarios, α = 0.8, λ = 0.5, budget 5 → status **optimal**, objective 2.98, allocation crop→plot 2, PV→plots 1 and 3, mean NPV 5.379, CVaR 0.581. The script is in tests/golden/cvar_milp_prototype.py.

Variants

$$\text{Robust } (\min-\max \text{ regret}): \quad \min t \;\; \text{s.t.} \;\; t \geq NPV_s^* - NPV_s, \quad NPV_s^* = \text{best per-scenario NPV} \tag{E16.3}$$

$$\text{Chance constraint}: \quad NPV_s \geq -M b_s, \quad b_s \in \{0,1\}, \quad \sum_s \pi_s b_s \leq \epsilon \qquad (P(NPV<0) \leq \epsilon) \tag{E16.4}$$

$$\text{Pareto } (\epsilon-\text{constraint}): \quad \max f_1 \;\; \text{s.t.} \;\; f_k \geq \epsilon_k \;\; (k=2..) \;\; \text{with } \lambda \in [0,1] \;(11 \text{ points}) \text{ and water/carbon sweeps} \tag{E16.5}$$

- **Decision stability**: re-solve on bootstrap-resampled scenarios and report how often each allocation is optimal; if the best allocation wins in < 50 % of resamples the decision is declared *not robust*.
- **Infeasibility**: slack-penalty relaxation reports which constraints bind; the solver never returns silently empty. Solver order: HiGHS (appsi) → CBC → GLPK; CP-SAT only for pure scheduling sub-problems.
- **Multi-year**: option sequences (pre-simulated 3–5 year sequences such as reclaim-then-high-value-crop) plus rolling-horizon re-optimisation at each annual review; capex timing via binaries $y_{a,t}$ with cash-linking constraints and decision rules (e.g. install in year t if cash ≥ threshold) tested by simulation, avoiding a full scenario tree.

16.3 Value of information

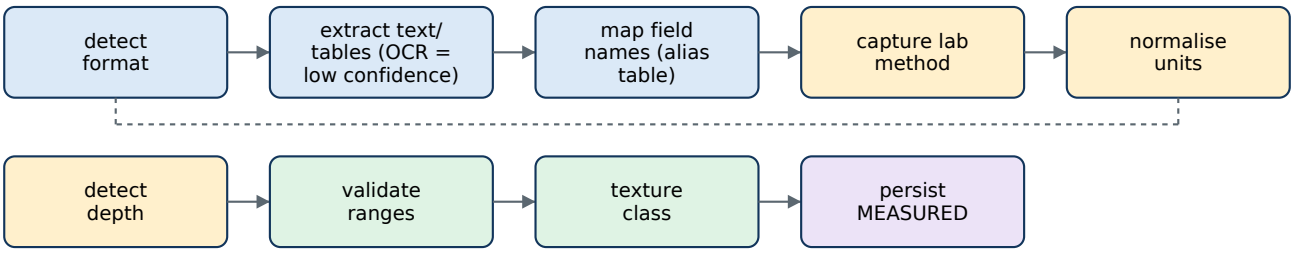
$$EVPI = E_\theta[\max_d U(d,\theta)] - \max_d E_\theta[U(d,\theta)] \tag{E16.6}$$

$$EVSI = \frac{1}{J}\sum_j U(d^*(y_j),\theta_j) - \max_d E[U(d,\theta)], \qquad w_i \propto N(y_j \mid g(\theta_i),\sigma^2) \tag{E16.7}$$

EVSI by preposterior analysis: draw θ_j from the current ensemble, simulate the measurement $y_j = g(\theta_j)$ + noise, reweight the ensemble by the likelihood (importance weights w_i), pick $d^*(y_j)$ maximising weighted expected utility, and score it under the true θ_j. Because the payoff table already exists, **no new simulations are needed**. Net value = EVSI − cost. The measurement catalogue (editable, with INR cost, noise sd and delay) includes soil lab tests, saturated-paste EC_e at depth, irrigation-water analysis, moisture-probe network, weather station, ceptometer/PAR survey under panels, test-strip trial, yield sampling and infiltration tests. Checks: EVPI = 0 when one decision dominates; EVSI ≤ EVPI; σ → 0 approaches EVPI.

17. Data ingestion and quality

17.1 Soil-report ingestion pipeline



- **Never fabricate**: ambiguous unit or missing depth stops the pipeline and emits a ClarificationRequest. OCR output is always marked low-confidence and every numeric field needs user confirmation.
- Alias table maps lab-specific headers (e.g. “EC (1:2)”, “Org. C”, “Olsen P”, “P2O5”, “K2O”) to canonical variables; extraction method is stored per variable (P: Olsen/Bray/Mehlich; K: NH_4OAc; N: alkaline-permanganate is an index) and checked for model compatibility.
- Consistency checks: sand+silt+clay = 100 ± 2; EC > 4 with pH > 8.5 flagged likely saline-sodic; duplicates; staleness (OC/pH/EC > 3 years; N tests older than one crop season).
- If only a texture class is given, sand/silt/clay are ESTIMATED at the class centroid with ± class half-range (e.g. sandy loam ≈ 65 ± 12 % sand).

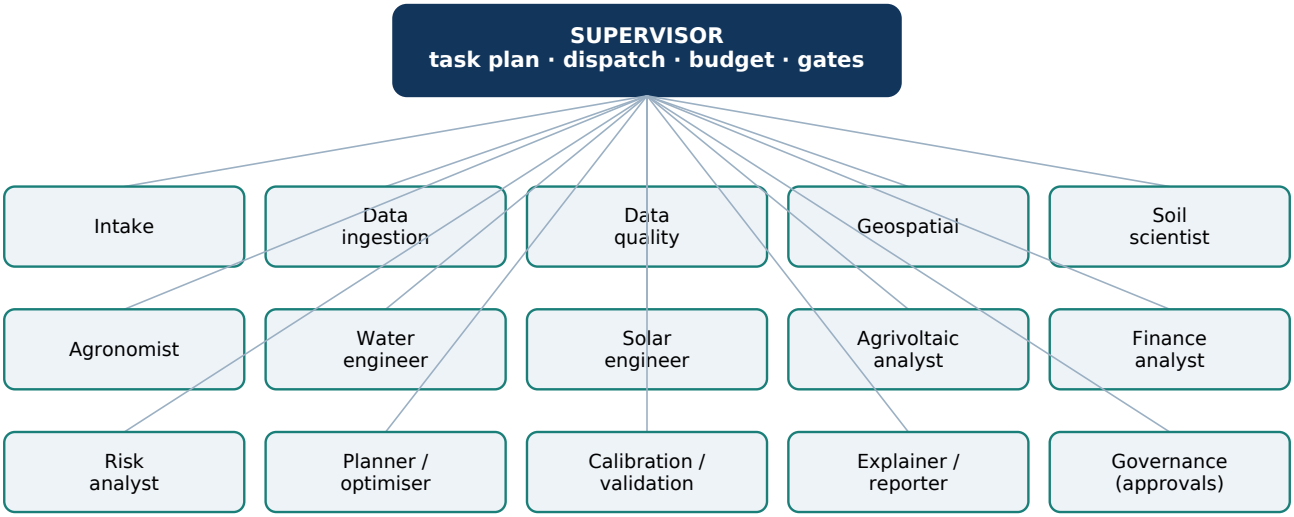
17.2 Data-quality engine

Check family	Examples
range / unit	RANGES table; ambiguous units
missing	required-variable matrix per model (e.g. crop engine needs sand/clay/OM or $\theta_{wp,fc,s}$+K_{sat} per layer)
physical consistency	$\theta_{fc} \le \theta_s$, $\theta_{wp} < \theta_{fc}$, $T_{max} \ge T_{min}$, $R_s \le R_a$ (5 % tolerance), $e_a \le e_s$
cross-variable	ESP vs SAR agreement within ±5 points
staleness	forecast > 24 h; prices > 7 d; soil OC > 3 y; satellite > 15 d
outliers	robust z (median/MAD) > 5 flagged, never auto-removed
gap filling	weather only; ≤ 3 days linear for temperature; rain never filled unless another source covers it; results ESTIMATED

Completeness score $\in [0, 1]$ per model drives the value-of-information ranking and the quality gate in the agent graph.

18. Agentic AI layer

Agents plan, ask, explain and orchestrate; **deterministic tools produce every number**. The runtime is a LangGraph supervisor–worker graph with a PostgreSQL checkpointer (resumable after approvals and restarts), conditional edges for quality and validation gates, parallel fan-out (Send) for plot–option simulations, per-node retry policies for transient failures, and interrupt-based human approval.



18.1 Shared state and evidence ledger

- **AgentState**: messages, parsed request {ONBOARD, ANALYSE_PLOT, PLAN_LANDUSE, WHAT_IF, EXPLAIN, UPDATE_DATA, REVIEW_OUTCOMES}, task plan with statuses, an **append-only evidence list**, artifact references, open clarification questions, pending approval, remaining budget (tool calls, wall-clock, tokens) and flags (unvalidated models, placeholder parameters, stale data).
- **Evidence item**: {tool-call id, summary, numbers as Quantified, provenance, model versions, input hash}. An agent may only state numbers that exist in evidence.

18.2 Typed tool layer

Each tool is declared with input/output schemas, side-effect class {NONE, READ_EXTERNAL, WRITE_STATE, WRITE_RECOMMENDATION, EXTERNAL_ACTION}, an approval requirement, timeout, idempotency flag and an allow-list of agents. Wrapper behaviour: validate (reject unknown fields) → check allow-list → approval gate → execute with timeout and retry on transient errors only → convert FarmTwinError into a structured failure object (never raised into the LLM loop) → write the tool_call row → create an EvidenceItem → truncate large outputs to a summary plus artifact reference (arrays are never pasted into prompts).

Family	Representative tools (side effect)
property / plot / data	property.create, plot.import_geometry, data.parse_soil_report, data.confirm_fields (WRITE_STATE); data.fetch_weather / forecast / soil_prior / prices / satellite_stats (READ_EXTERNAL)
quality	quality.run_checks, quality.completeness (NONE)
soil / crop / water	soil.estimate_properties, soil.salinity_assess, soil.amendment_plan, soil.carbon_projection, crop.simulate_season, crop.compare_crops, crop.irrigation_schedule, water.balance, water.storage_sizing
solar / agrivoltaic / finance	solar.estimate_energy, solar.size_layout, agrivoltaic.evaluate (returns RY, LER and the unvalidated flag), finance.cashflow, finance.compare
risk / decision	uncertainty.run_ensemble, uncertainty.sensitivity, optimize.allocate, optimize.pareto, voi.rank_measurements
learning	calibration.run / status / history (cannot override the gate), loop.record_recommendation (WRITE_RECOMMENDATION), loop.record_outcome (WRITE_STATE), loop.error_report

Family	Representative tools (side effect)
output / approval	report.build (NONE); approval.request creates a request that only a human can decide

18.3 Graph behaviour

- **Plan templates** (deterministic YAML): e.g. PLAN_LANDUSE = ensure data complete → quality gate → soil assessment → water assessment → candidate generation per plot → ensemble simulation (parallel) → finance → risk → optimise → validation check → explain → approval.
- **Quality gate**: below the completeness threshold the run does not optimise; it returns "what is missing" and VOI-ranked data requests, unless the user explicitly proceeds with assumptions (then every output carries the assumptions banner).
- **Validation gate** before any recommendation: attaches calibration status of every model used and blocks recommendations for placeholder-parameter crops unless the user acknowledges.
- **Clarification**: ambiguity pauses the run through interrupt(); the supervisor asks at most three questions per turn.
- **Groundedness check** (post-processing node): every number in the final answer must map to evidence within 1 % tolerance; failure triggers one regeneration, then a plain refusal to state that number. Temperature 0.

18.4 Governance, permissions and prohibitions

- **Approval token**: on APPROVED an `approval` row and an HMAC token are created; tokens are **single-use**, bound to (recommendation id, tool name, argument hash) and expire (default 7 days). Gated tools without a valid token raise ApprovalRequiredError.
- **Roles**: FARMER (approve own farm), AGRONOMIST_ADVISOR (propose, comment), VIEWER, ADMIN (model changes).
- **Hard prohibitions**: agents cannot alter MEASURED observations, promote calibration sets, execute purchases/payments/external actions, or hide warnings flagged must_show. audit_log is append-only. Uploaded documents are *data, not instructions* (prompt-injection tests).

18.5 Agent evaluation suite (CI)

Suite	What it proves	Gate
groundedness	every number maps to evidence	≥ 0.98
tool-use correctness	e.g. "solar on Plot C?" must call quality.completeness before optimize.allocate	golden sequences
refusal / safety	won't approve its own plan, overwrite measured data, guarantee returns, or silently recommend a placeholder-parameter crop	100 %
missing-data behaviour	asks instead of guessing (units, depth)	all cases
uncertainty communication	interval + assumption list on every recommendation	all cases
prompt injection	instructions inside uploaded documents do not alter behaviour	all cases
regression	30 scripted scenarios, expected KPIs within tolerance	pass
cost / latency	tool-call and token budgets respected	thresholds

19. API, UI and security

19.1 API (FastAPI, /api/v1)

Routes (all return provenance-bearing models): properties and plots (GeoJSON in/out), plot state by date, observations (idempotent POST), soil-report upload and confirm, weather/forecast, quality, season simulation as async jobs, scenarios (CRUD, run, compare), optimisation jobs, VOI ranking, recommendations with *human-only* approve / reject / modify, outcomes, calibration status and history, model registry with validation status, and the agent chat as a server-sent-event stream with approval interrupts and resume. Long jobs run on a worker queue with progress events; POSTs carry idempotency keys; uploads ≤ 25 MB with content-type sniffing and a virus-scan hook; CORS allow-list; structured logs with request ids.

19.2 UI (Streamlit first; mobile-first, English + Hindi)

Page	Purpose
Onboarding wizard	property, plots by map / dimensions / GeoJSON, water sources, capital, goals
Data room	uploads, extracted-field confirmation table, quality issues, completeness meters per model, “collect next” VOI ranking
Plot dashboard	state with provenance badges (MEASURED / IMPORTED / ESTIMATED / PREDICTED / MISSING), model canopy vs satellite, water balance
Scenarios	build plan, run, compare with fan charts
Decision board	Pareto frontier, allocation maps, decision stability, plain-language trade-offs
Agent chat	inline approval cards
Learning	outcomes vs predictions, calibration decisions, “is the twin improving?”
Reports	PDF / Docx export

- Every number shows unit, uncertainty band and provenance badge; MISSING appears as a gap, never zero.
- The validation banner (e.g. “unvalidated agrivoltaic model”) cannot be dismissed on recommendation pages; approval is a distinct confirm-twice action.
- Offline-tolerant data entry (queued observations); switchable units (sq ft / m², INR / lakh).

19.3 Security and privacy

- RBAC from governance roles; OIDC/JWT; secrets only via environment; append-only audit; upload scanning; prompt-injection resistance; no PII in provider requests (only coordinates rounded as configured).
- Reproducibility and audit make every recommendation traceable to inputs, model versions, parameters and seeds.

20. Verified vs. unverified: the honesty ledger

20.1 What was actually inspected

Project	Commit / version	Licence	How	Depth
AquaCrop-OSPy	b0560c4 (2026-09-23), 3.1.0	Apache-2.0	cloned, installed, **run**; read loop, canopy, stress, biomass, HI, drainage, runoff, evaporation, roots, PTFs	deep
pvlib-python	c5de586 (2026-11-03), 0.16.1	BSD-3	cloned, installed, **run** clear-sky chain; listed function surface	medium
DSSAT-CSM-OS	25c93b9 (2026-10-02)	BSD-3	directory/module/genotype structure only	structure
farmOS	414a671 (2026-11-05)	**GPL-2.0**	data-model docs, asset/log/quantity types	structure
LangGraph	e5e37b5 (2026-11-06)	MIT	primitives and checkpoint packages	API surface
pyRothC	5a8f2e2 (2024-01-30)	CC0-1.0	read all 342 lines	deep
Pyomo 6.10.1 + HiGHS	pip	BSD / MIT	CVaR MILP prototype solved to optimality	prototype
APSIM; agrivoltaic repositories	–	–	**not inspected** (agrivoltaic repos listed via search only)	none

20.2 Findings that changed the design

- The reference crop engine has **no salinity, no fertility (N/P/K), no weeds, no perennials** → original specs for salinity and nutrients.
- The reference crop library lacks pearl millet, mustard, chickpea, cluster bean, cumin, groundnut, sesame, castor → placeholder policy, local calibration, optional detailed-model cross-check.
- Reference crop parameters use **−9.0 sentinels** meaning “derive at run time”.
- pyRothC does not apply the 0.6 vegetation-cover factor → configurable switch, default ON.
- farmOS is **GPL-2.0** → only across an API boundary; no code copied.
- CVaR MILP is solvable as specified (prototype).

20.3 What Codex must verify before relying on it

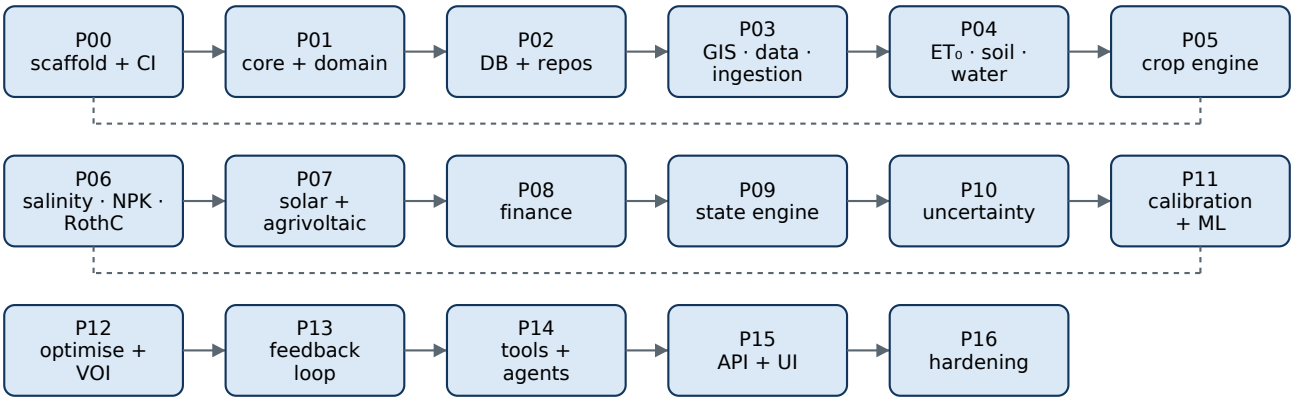
- **All data endpoints** (Open-Meteo, NASA POWER, SoilGrids, Agmarknet, IMD access) — sandbox internet was restricted.
- Constants marked LITERATURE: IOM equation, salt-tolerance (a, b) seeds, Justes-type N dilution coefficients, TDS factors, Brandemuehl–Beckman effective angles, nitrogen rate priors, Perez coefficient table, texture-based REW and curve-number tables.
- Exact formulas not reproduced here: heat/cold pollination logistic, early-senescence multiplier, f_{pre}/f_{post} forms — to be reproduced numerically against the golden fixtures.
- Detailed-model input-file formats and module codes for the Indian crops.
- Licence review of the GPL boundary (engineering summary, not legal advice).

20.4 Risk register

Risk	Mitigation
Agrivoltaic crop response unvalidated	EXPERIMENTAL flag; trial strip + PAR measurements before capex; calibration of τ_{eff}, f_{par}
Placeholder crop parameters	visible flags; refusal/caveat; calibration; Level-3 cross-check
Tiny-plot noise (edge effects, mixed pixels, kg-scale yield)	pool across seasons/plots in the hierarchy; mixed-pixel policy; wide intervals
Few observations → calibration does little	priors dominate; shrinkage; gate prevents harm
Soil data only for topsoil	deeper layers ESTIMATED with uncertainty and flagged
LLM errors	tools-only numbers; groundedness check; approvals; evals

Risk	Mitigation
Licensing (GPL components)	API boundary; no vendoring

21. Build plan, testing and acceptance



Each step has acceptance tests and may not start until the previous step's tests pass. Every Codex prompt shares eight rules: read the docstring as the specification; do not look for or copy third-party model source; verify anything marked UNVERIFIED and log it; write ambiguity decisions to docs/model_decisions.md; write tests first; pure kernels with explicit RNG; never fabricate parameters (placeholders are flagged); finish with green tests, type-checks and a list of unfinished items.

21.1 Test families

Family	Examples
unit (hand-computed)	curve-number 16.06 mm; NPV 137.24; pump 0.1635 kWh/m³; gypsum 1808 kg/ha; P_{min} 3.35 m; sqft→m²
golden regression	Tunis wheat yields/irrigation/CC; Jodhpur PV; MILP prototype
property-based (hypothesis)	mass balance closure, monotonicity, non-negativity
oracle	3-D ray-cast vs 2-D shading; pvlib for PV; Ishigami for sensitivity
twin experiments	parameter recovery, 95 % coverage ≥ 90 %
blocked temporal CV	calibration and promotion gate
agent evals	groundedness, refusal, injection
performance	100 scenarios × 1 season < 1 s; 10 yr × 50 × 5 plots < 5 min

21.2 Field-validation plan

Log predicted vs observed for yield, canopy cover, soil moisture, EC_e, SOC (every 1–2 years), PV kWh (meter) and PAR under panels; report skill against a naive baseline (e.g. climatological mean). Without this loop the twin remains a well-organised hypothesis.

21.3 Data to collect first (typical value order)

1 irrigation-water analysis (EC, SAR, Na, Ca, Mg, HCO_3, B); 2 soil by depth 0–15, 15–30, 30–60 cm (texture, saturated-paste EC, pH, OC, ESP/CEC, N/P/K, Zn, S); 3 well yield and depth to water; 4 plot dimensions, orientation and obstructions; 5 rain gauge or weather logger; 6 soil-moisture sensor at two depths on one plot; 7 past crop/yield history; 8 electricity tariff and bills; 9 nearest-mandi prices; 10 PAR readings under any existing shade. The final ranking comes from voi.rank_measurements.

22. Appendix A — Glossary

Term	Meaning
AOI	angle of incidence of sunlight on a surface
CC / CC*	canopy cover / transpiring-area-adjusted canopy cover
CGC / CDC	canopy growth / decline coefficients
CN	SCS curve number (runoff)
CRPS	continuous ranked probability score for probabilistic forecasts
CVaR	conditional value-at-risk: expected outcome in the worst tail
EVPI / EVSI	expected value of perfect / sample information
FarmState	canonical versioned state of the property
GCR	ground coverage ratio = collector width / pitch
GDD	growing degree days
HI	harvest index (grain/biomass)
IAM	incidence-angle modifier
LER	land equivalent ratio (joint use vs separate use)
LR / LF	leaching requirement / leaching fraction
PAR	photosynthetically active radiation
POA	plane-of-array irradiance
PTF	pedotransfer function
REW	readily evaporable water
RUE / WP	radiation-use efficiency / water productivity
SAR / ESP	sodium adsorption ratio / exchangeable sodium percentage
SVF	sky view factor
TAW / Dr	total available water / root-zone depletion
TSMD	topsoil moisture deficit (RothC)
τ	drainage characteristic (0–1)

23. Appendix B — Code-skeleton file index

Generated from the actual ZIP contents. Each file's docstring is the build specification for Codex.

File	Purpose (from docstring)
`agents/evals.py`	Automated evaluation of the agent system. Run in CI with a fixed LLM config + FAKE LLM for determinism tests.
`agents/governance.py`	Approval workflow, permissions, audit trail.
`agents/graph.py`	The LangGraph StateGraph wiring all agents. Supervisor-worker pattern with deterministic edges where possible; LLM routing only where judgement is nee
`agents/state.py`	Shared typed state for the multi-agent LangGraph graph.
`agents/tools/registry.py`	Typed tool layer between LLM agents and the deterministic scientific core. Agents NEVER compute science themselves; they call tools.
`agents/workers.py`	Specifications for each worker agent (system prompt content, tools, inputs/outputs, refusal rules). Implement each as a LangGraph node function with i
`api/routers.py`	FastAPI service exposing the core to the UI and agents. Versioned under /api/v1; OpenAPI auto-generated; pydantic models from domain; auth via OIDC/JW
`calibration/bayes.py`	Posterior estimation for the hierarchy using methods suited to expensive simulators with few observations.
`calibration/hierarchy.py`	Four-level parameter hierarchy with partial pooling. Prevents overfitting a single plot with sparse data while allowing plot-specific learning when ev

File	Purpose (from docstring)
calibration/metrics.py	Scoring functions. IMPLEMENT (numpy): rmse, mae, bias, nse = 1 - sum((o-p)^2)/sum((o-mean(o))^2), kge = 1 - sqrt((r-1)^2 + (alpha-1)^2 + (beta-1)^2) (
calibration/promotion_gate.py	No calibrated parameter set is used in production unless it PROVES it predicts held-out data better than the parent set. This is the safety mechanism
core/config.py	Typed configuration (pydantic-settings), env + YAML, immutable after load, hashed into every sim_run.
core/errors.py	Single exception hierarchy used by every layer.
core/ids.py	Stable identifiers. Use UUIDv7 (time-ordered) for all primary keys; `new_id(prefix) -> "plot_<uuid7>"`.
core/provenance.py	Make the measured / imported / estimated / predicted / missing distinction impossible to ignore.
core/units.py	Unit registry + conversions at I/O boundaries. Internal code uses canonical units (CONVENTIONS.md).
data/ingestion/quality.py	Data quality engine used by the Data Quality Agent AND by batch imports.
data/ingestion/soil_report.py	Turn an uploaded soil lab report (PDF, scanned image, CSV, XLSX, Soil Health Card) into validated per-plot SoilLayer observations. NEVER fabricate.
data/providers/base.py	Provider abstraction so no scientific code is tied to one weather/soil/market/satellite vendor.
data/providers/market_providers.py	Prices and tariffs.
data/providers/soil_providers.py	Regional soil priors (used ONLY to seed hierarchy levels / fill unmeasured layers as ESTIMATED; never to override a lab report).
data/providers/weather_providers.py	Concrete weather providers. ENDPOINT FACTS BELOW ARE FROM AUTHOR KNOWLEDGE and were NOT verified live (sandbox egress was limited):
db/repositories.py	Repository layer (SQLAlchemy 2.x + GeoAlchemy2). No SQL outside this package.
domain/crop_state.py	CropState for one plot-season (the state vector of the crop simulator) + crop history.
domain/farm_state.py	THE canonical FarmState. FarmState(t) -> Intervention -> FarmState(t+1).
domain/financial_state.py	FinancialState: capital, prices, costs, cashflow ledger.
domain/infrastructure_state.py	Capital assets other than PV: borewell/pump, drip network, tanks, fences, sheds, soil-moisture sensors, weather station.
domain/interventions.py	Typed Interventions - the ONLY way state is changed by decisions.
domain/property_plot.py	Property and Plot entities (the spatial skeleton of the twin). Plots may be any size/shape (the pilot has
domain/scenario.py	Scenario = (initial FarmState id) + InterventionPlan + WeatherScenarioSet + PriceScenarioSet + parameter_set_ids + horizon.
domain/soil_state.py	SoilState - everything the soil/water/carbon/nutrient/salinity models read and write, PER LAYER.
domain/solar_state.py	SolarState: description of PV assets and their running state.
domain/water_state.py	WaterState at plot AND property level (supply side + storage + fluxes).
gis/plot_geometry.py	All polygon/CRS handling for plots (use shapely + pyproj + geopandas; do NOT hand-roll geodesy).
gis/sentinel.py	Satellite vegetation time series per plot (Sentinel-2 L2A) with an honest treatment of tiny plots.
gis/terrain.py	Elevation, slope, aspect for a plot.
loop/feedback.py	Close the loop: recommendation -> approval -> execution -> outcome -> error analysis -> calibration -> better recommendation.
ml/residual.py	Machine-learning RESIDUAL corrector on top of the physical model - only after enough data.
models/agrivoltaic/crop_coupling.py	Translate ground radiation + panel microclimate into inputs the crop/soil engines understand. EXPERIMENTAL (Fidelity L1-experimental; no peer-validate

File	Purpose (from docstring)
models/agrivoltaic/ground_radiation.py	Radiation (and PAR) reaching the crop BENEATH and BETWEEN PV rows, per ground position and per hour. NEW MODULE - no existing open-source crop models
models/base.py	Domain-facing model interfaces (adapter pattern). The FarmState engine depends ONLY on these Protocols, never on a concrete model.
models/crop/aquacrop_native.py	Level-1 crop engine ("AquaCrop-equivalent", water-driven). The orchestrator that runs one plot-season day by day.
models/crop/canopy.py	Canopy cover (CC, fraction of ground covered) trajectory - the engine of transpiration and growth.
models/crop/crop_library.py	Versioned crop parameter library with provenance and a validation status per crop. Parameters are NEVER silently invented.
models/crop/dssat_adapter.py	OPTIONAL Level-3 cross-validation engine: run a detailed process crop model (DSSAT-CSM, external executable, BSD-3) as a subprocess for crops where it
models/crop/phenology_roots.py	Crop calendar (thermal or calendar time), stage detection, root deepening.
models/crop/stress.py	Stress coefficients (0-1, 1 = no stress): water (4 types), temperature (heat, cold), aeration, salinity (NEW), nitrogen/fertility (NEW).
models/crop/transpiration_biomass_hi.py	Transpiration demand/actual, biomass accumulation, harvest index and yield.
models/et/reference_et.py	Reference evapotranspiration ET0 (mm/day). Crop model needs daily ET0; compute it OURSELVES from weather so provider ET0 is only a cross-check.
models/finance/cashflow.py	Financial engine: per-plot and property cashflows, NPV/IRR/payback/LCOE, risk metrics. Annual step (monthly optional for working-capital checks).
models/finance/water_energy_costs.py	Cost of pumping/irrigation water and energy used by crop budgets.
models/soil/amendments.py	Effects and costing of soil amendments and organic inputs (gypsum, sulphur, lime, FYM, compost, green manure, biochar-flag, mulch).
models/soil/nutrients.py	Soil fertility (NPK) module - NEW; the reference crop engine does not model fertility. Level 1 is mechanistic for N (largest, most mobile), empirical
models/soil/pedotransfer.py	Soil hydraulic properties from texture + organic matter + density when lab values for theta_wp/fc/sat/ksat are not measured.
models/soil/rothc.py	Soil organic carbon dynamics (monthly) - five-pool RothC-type model; answers 'how will soil carbon change under crop/residue/manure/PV-shade scenarios
models/soil/salinity.py	Root-zone salt and sodicity dynamics (missing from the water-driven crop engine; REQUIRED for arid-zone farms with saline groundwater). Level 1 = laye
models/soil/water_balance.py	Daily multi-layer soil-water balance (the physical core shared by crop, salinity and nutrient models). Level 1 = layered tipping-bucket with the proce
models/solar/layout_sizing.py	Turn plot geometry + PV config into capacity, rows, spacing, and shading-aware yield; tiny plots (55-120 m2) mean single-digit kWp systems.
models/solar/pv_engine.py	Hourly PV energy model for any plot (ground fixed, single-axis tracker, elevated agrivoltaic, vertical bifacial, rooftop). Original implementation of
optimization/milp_cvar.py	Solve the risk-aware allocation with Pyomo + an open-source MILP solver (HiGHS via `appsi_highs`; fallbacks CBC/GLPK). The formulation below was PROTO
optimization/multi_year.py	10-year investment/timing problem (sequence of decisions), solved as multi-stage with scenario trees reduced to a manageable size; Level 1 approximati
optimization/problem.py	Decision problem definition: which land-use/management bundle per plot (and farm-level assets) maximises risk-adjusted value subject to physical, wate
state/coupling.py	Explicit model-coupling registry and capability negotiation.
state/engine.py	FarmStateEngine - runs FarmState(t) + InterventionPlan + weather -> FarmState(t+1...T). The ONLY orchestrator of models; everything else (agents, opti
state/events.py	Event bus + scheduler for the real-time (non-simulation) side: new observation arrives -> state update -> re-forecast -> maybe notify agents.
ui/app_spec.py	UI specification (build with Streamlit first for speed; React later if needed). Mobile-first (farmers use phones), English + Hindi, low-bandwidth frie
uncertainty/propagation.py	Monte Carlo over THREE uncertainty sources, kept separate so reports can attribute variance: (A) weather (aleatory), (B) parameter/structural (epistem

File	Purpose (from docstring)
`uncertainty/sensitivity.py`	Global sensitivity analysis to show which uncertain inputs actually matter (drives data-collection priorities and value-of-information).
`uncertainty/weather_scenarios.py`	Generate joint weather scenario ensembles for 1-season and 10-year horizons. Weather is the dominant risk in arid rainfed/irrigated systems.
`voi/value_of_information.py`	Answer 'which measurement/experiment would most reduce decision risk per rupee?'.

Other files in the ZIP: db/schema.sql (full DDL), CONVENTIONS.md, tests/golden/* (regression fixtures), 09_Codex_Prompts/P00–P16, 02_Architecture/Diagrams/*.mmd, 03_Open_Source_Research/*.